Group Optimization Report

Objective (aggregated fitness): 0.0272623
Penalty (total aggregated): 0
Algorithm: Classic + Nelder–Mead (final polish)
Population size: 384
Parameter vector: 32 params (shared[0-10], Bas_2[11-17], Bas_1[18-24], Bas_0[25-31])
Input file: propellants.01234.json
Generated: 2026-08-07 03:16

Run Configuration (as resolved)

Provenance

- **Mode:** optimization
- **Configuration source:** /root/runs/6h-Classic-Tm2300-kc400-constrON-fsKinetic-20260806-2106/run-config.json
- **Working directory:** /root/runs/6h-Classic-Tm2300-kc400-constrON-fsKinetic-20260806-2106
- **Input propellants file:** propellants.01234.json
- **Resolved sidecar:** run_configuration.resolved.json
- **Recorded at (UTC):** 2026-08-06 21:07:05

Model constants

- **Metal melting temperature, K:** 2300
- **Surface-temperature search bracket, K:** 600 .. 900
- **Skeleton conduction contact factor:** 400 (contact-area fraction 0.0025; fixed calibration, not fitted)
- **Skeleton surface fraction:** kinetic coverage: $f_s = 1/(1 + a_0 + a_1 \cdot w_{fine} \cdot (p/p_{ref})^m)$ — accumulation against burnout of the binder residue; independent of the layer thickness

burnout channels: a_0 = 0.9966 (binder), a_1 = 3.3653 (fine oxidiser), m = 0.71 at p_{ref} = 1 MPa

calibration: 3 shared constants fitted on Bas_2/Bas_3/Bas_4 (one binder, one additive, AP dispersity varied); 4.7–8.6 % RMS in-sample, 12.7–20.9 % leaving one composition out. NOT calibrated for Bas_0 or Bas_1 — different binder

Search

- **Strategy:** Classic
- **Parameter vector dimension:** 32
- **Population size (effective):** 384
- **Worker processors (effective):** 32 of 36 machine cores
- **RNG seed:** 20260729 (reproducible at 32 workers)
- **Termination:** stagnation streak 2,147,483,647 @ rel 1E-06 OR 6 h safety timeout
- **Mutation force F:** 0.5
- **Crossover probability CR:** 0.9

Nelder-Mead refinement

- **Enabled:** yes
- **Stages:** final polish (100,000 evals)
- **Adaptive coefficients:** yes
- **Domain tolerance:** 1E-08
- **Function tolerance:** 1E-08
- **Restarts:** 2

Penalty thresholds

- **Penalty rate:** 0.01
- **Pocket heat-flux ratio threshold:** 100
- **Pore diameter threshold:** 3
- **Large oxidiser particle size threshold:** 1
- **Max inter-pocket kinetic-flame heat flux, W/m2:** 1E+09
- **Max skeleton kinetic-flame heat flux, W/m2:** 1E+08
- **Max out-skeleton kinetic-flame heat flux, W/m2:** 1E+08

Parameter bounds (18 base parameters)

ADecompose: [1, 1E+15]
EDecompose: [50000, 300000]
AKineticFlameInterPocket: [100000, 1E+13]
EKineticFlameInterPocket: [50000, 250000]
AKineticFlamePocketOutSkeleton: [100000, 1E+13]
EKineticFlamePocketOutSkeleton: [50000, 250000]
AKineticFlamePocketSkeleton: [100000, 1E+13]
EKineticFlamePocketSkeleton: [50000, 250000]
NuInterPocket: [0, 2.5]
NuPocketOutSkeleton: [0, 2.5]
NuPocketSkeleton: [0, 2.5]
AMetalBurningConstant: [1E-10, 0.001]
BMetalBurningConstant: [1E-12, 1E-05]
DeltaH: [-1E+07, 1E+07]
KDiffusionHeight: [0.001, 10]
APowOrder: [0, 3]
BPowOrder: [0, 3]
KCoefficientRadiationTemperature: [0, 1]

Functional Constraints

Flame competition constraint within pocket regions.
Penalty coefficient 0.0100
Maximum allowable heat flux density ratio (max/min) 100.00
The linear burning rate of the first conditional fuel is greater than that of the second.
Penalty coefficient 0.0100
Constraint on maximum kinetic flame heat flux density.
Penalty coefficient 0.0100
Maximum kinetic flame heat flux density in inter-pocket medium 1.00e+09
Maximum kinetic flame heat flux density in skeleton structure 1.00e+08
Maximum kinetic flame heat flux density in pocket region outside skeleton 1.00e+08
Constraint on skeleton thickness to pore diameter ratio.
Penalty coefficient 0.0100
Threshold for skeleton thickness to pore diameter ratio 3.00
Constraint on skeleton thickness to large oxidizer particle diameter ratio.
Penalty coefficient 0.0100
Threshold for skeleton thickness to large oxidizer particle diameter ratio 1.00

Parametric Constraints

Constraint on allowable surface temperature range
Minimum surface temperature 600 K
Maximum surface temperature 900 K

Differential Evolution Algorithm

Population size: 384
Mutation force: 0.5
Crossover probability: 0.9
Number of threads: 32
Termination strategy: OrTerminationStrategy

Group Combustion Solver Parameters

Values below are display-rounded — not for replay; use the Reproduction Vector block.
Each parameter shows its bounds and a rail flag: [RAILED@lo]/[RAILED@hi] means the gene sits within 0.1% of a bound (fit is bound-limited — consider widening); [interior] means it does not.

<u>Shared Parameters (All Compositions)</u>

Pre-exponential coefficient for gas phase reactions in inter-pocket medium 1954275689.1199553
Parameter bounds [1.000E+005; 1.000E+013] [interior]

Activation energy for gas phase reactions in inter-pocket medium 206425.68046919547
Parameter bounds [5.000E+004; 2.500E+005] [interior]
Pre-exponential factor for gas-phase reactions in pocket region outside skeleton 1176884.484228617
Parameter bounds [1.000E+005; 1.000E+013] [interior]
Activation energy for gas-phase reactions in pocket region outside skeleton 50000.35358104978
Parameter bounds [5.000E+004; 2.500E+005] [RAILED@lo]
Reaction order for gas-phase chemistry in inter-pocket medium 2.4999990002476715
Parameter bounds [0.000E+000; 2.500E+000] [RAILED@hi]
Reaction order for gas-phase chemistry outside skeleton structure 1.1193352413556008
Parameter bounds [0.000E+000; 2.500E+000] [interior]
Enthalpy difference between vaporization and binder decomposition -407243.6375001921
Parameter bounds [-1.000E+007; 1.000E+007] [interior]
Diffusion flame height correlation coefficient 1.0990
Parameter bounds [1.000E-003; 1.000E+001] [interior]
Power law exponent for burning rate dependency on A (skeleton thickness) 1.3241
Parameter bounds [0.000E+000; 3.000E+000] [interior]
Power law exponent for burning rate dependency on B (pore diameter) 0.0761
Parameter bounds [0.000E+000; 3.000E+000] [interior]
Radiation temperature correlation coefficient (average layer temperature) 0.9457
Parameter bounds [0.000E+000; 1.000E+000] [interior]

Bas_2 (includes Bas_3, Bas_4) - Condensed Phase Parameters
Pre-exponential coefficient for condensed phase reactions 997662548476933.5
Parameter bounds [1.000E+000; 1.000E+015] [interior]
Activation energy for condensed phase reactions 162857.19191489473
Parameter bounds [5.000E+004; 3.000E+005] [interior]
Pre-exponential factor for gas-phase reactions within skeleton structure 298661.8953744815
Parameter bounds [1.000E+005; 1.000E+013] [interior]
Activation energy for gas-phase reactions within skeleton structure 50000.14901279834
Parameter bounds [5.000E+004; 2.500E+005] [RAILED@lo]
Reaction order for gas-phase chemistry in skeleton structure 2.1966089278159577
Parameter bounds [0.000E+000; 2.500E+000] [interior]
Coefficient A for metal combustion heat flux density in skeleton structure, 5.372748267231791E-08
Parameter bounds [1.000E-010; 1.000E-003] [RAILED@lo]
Coefficient B for metal combustion heat flux density in skeleton structure, 1.3351465953259384E-06
Parameter bounds [1.000E-012; 1.000E-005] [RAILED@lo]

Bas_1 - Condensed Phase Parameters
Pre-exponential coefficient for condensed phase reactions 999952190086927.5
Parameter bounds [1.000E+000; 1.000E+015] [RAILED@hi]
Activation energy for condensed phase reactions 165077.49442030492
Parameter bounds [5.000E+004; 3.000E+005] [interior]
Pre-exponential factor for gas-phase reactions within skeleton structure 5664676192726.616
Parameter bounds [1.000E+005; 1.000E+013] [interior]
Activation energy for gas-phase reactions within skeleton structure 200832.92962550418
Parameter bounds [5.000E+004; 2.500E+005] [interior]
Reaction order for gas-phase chemistry in skeleton structure 2.2263981667869626
Parameter bounds [0.000E+000; 2.500E+000] [interior]
Coefficient A for metal combustion heat flux density in skeleton structure, 3.7730960428959604E-09
Parameter bounds [1.000E-010; 1.000E-003] [RAILED@lo]
Coefficient B for metal combustion heat flux density in skeleton structure, 1.2277953178129537E-07
Parameter bounds [1.000E-012; 1.000E-005] [RAILED@lo]

Bas_0 - Condensed Phase Parameters
Pre-exponential coefficient for condensed phase reactions 997854333602391.9
Parameter bounds [1.000E+000; 1.000E+015] [interior]
Activation energy for condensed phase reactions 224096.5432749049
Parameter bounds [5.000E+004; 3.000E+005] [interior]
Pre-exponential factor for gas-phase reactions within skeleton structure 8049832814810.603

Parameter bounds [1.000E+005; 1.000E+013] [interior]
Activation energy for gas-phase reactions within skeleton structure 227484.92376765012
Parameter bounds [5.000E+004; 2.500E+005] [interior]
Reaction order for gas-phase chemistry in skeleton structure 0.7233727430915838
Parameter bounds [0.000E+000; 2.500E+000] [interior]
Coefficient A for metal combustion heat flux density in skeleton structure, 2.7660575274175673E-08
Parameter bounds [1.000E-010; 1.000E-003] [RAILED@lo]
Coefficient B for metal combustion heat flux density in skeleton structure, 1.1397592154883802E-06
Parameter bounds [1.000E-012; 1.000E-005] [RAILED@lo]

Reproduction Vector

Full-precision result vector. Copy the block below (including the '#' lines) into a text file and replay this run exactly with: --forward-eval <file>

reproduction vector (32 params)
objective (aggregated fitness) = 0.027262267849997728
penalty (total aggregated) = 0
input file = propellants.01234.json
population = 384
generated = 2026-08-07 03:16
layout = shared[0-10], Bas_2[11-17], Bas_1[18-24], Bas_0[25-31]
count = 32
checksum = 91c54097d9232ae24f18b41fb630cb2ee9e19280141afb3ed7ce9f824697e8e4
1954275689.1199553
206425.68046919547
1176884.484228617
50000.35358104978
2.4999990002476715
1.1193352413556008
-407243.6375001921
1.0989591917473231
1.3240910867110078
0.07612057549109397
0.9456825008515037
997662548476933.5
162857.19191489473
298661.8953744815
50000.14901279834
2.1966089278159577
5.372748267231791E-08
1.3351465953259384E-06
999952190086927.5
165077.49442030492
5664676192726.616
200832.92962550418
2.2263981667869626
3.7730960428959604E-09
1.2277953178129537E-07
997854333602391.9
224096.5432749049
8049832814810.603
227484.92376765012
0.7233727430915838
2.7660575274175673E-08
1.1397592154883802E-06

Optimization Results

Objective function value 0.0273

Total penalty from constraint violations 0.0000

Burn-Rate Error Breakdown

*Per-point relative error e = (calc/exp - 1)*100%. A fuel's RMS is its contribution to the objective (sqrt of the mean squared fractional error over all pressures); a composition objective is the mean of its per-fuel RMS, and the overall objective is the mean of the three compositions. Numbers below reflect this report's input vector.*

Bas_2 (includes Bas_3, Bas_4) - Burn-Rate Errors

Bas_2

p = 1 MPa: exp 16.9441, calc 14.3591 mm/s -> err -15.26%
p = 1.611 MPa: exp 21.7132, calc 18.7756 mm/s -> err -13.53%
p = 2.222 MPa: exp 25.6654, calc 22.7347 mm/s -> err -11.42%
p = 2.833 MPa: exp 29.1215, calc 26.4111 mm/s -> err -9.31%
p = 3.444 MPa: exp 32.2345, calc 29.8844 mm/s -> err -7.29%
p = 4.056 MPa: exp 35.0917, calc 33.1996 mm/s -> err -5.39%
p = 4.667 MPa: exp 37.7487, calc 36.3860 mm/s -> err -3.61%
p = 5.278 MPa: exp 40.2432, calc 39.4640 mm/s -> err -1.94%
p = 5.889 MPa: exp 42.6026, calc 42.4484 mm/s -> err -0.36%
p = 6.5 MPa: exp 44.8470, calc 45.3508 mm/s -> err +1.12%
Per-fuel RMS = 0.0856 (8.56%)

Bas_3

p = 1 MPa: exp 15.2411, calc 15.5700 mm/s -> err +2.16%
p = 1.611 MPa: exp 20.6812, calc 20.6692 mm/s -> err -0.06%
p = 2.222 MPa: exp 25.4073, calc 25.1918 mm/s -> err -0.85%
p = 2.833 MPa: exp 29.6815, calc 29.4192 mm/s -> err -0.88%
p = 3.444 MPa: exp 33.6334, calc 33.4467 mm/s -> err -0.56%
p = 4.056 MPa: exp 37.3394, calc 37.3182 mm/s -> err -0.06%
p = 4.667 MPa: exp 40.8488, calc 41.0590 mm/s -> err +0.51%
p = 5.278 MPa: exp 44.1961, calc 44.6867 mm/s -> err +1.11%
p = 5.889 MPa: exp 47.4063, calc 48.2141 mm/s -> err +1.70%
p = 6.5 MPa: exp 50.4986, calc 51.6516 mm/s -> err +2.28%
Per-fuel RMS = 0.0127 (1.27%)

Bas_4

p = 1 MPa: exp 9.7279, calc 11.3748 mm/s -> err +16.93%
p = 1.611 MPa: exp 13.5834, calc 13.8856 mm/s -> err +2.23%
p = 2.222 MPa: exp 17.0126, calc 16.4880 mm/s -> err -3.08%
p = 2.833 MPa: exp 20.1664, calc 19.2386 mm/s -> err -4.60%
p = 3.444 MPa: exp 23.1208, calc 22.0990 mm/s -> err -4.42%
p = 4.056 MPa: exp 25.9212, calc 25.0333 mm/s -> err -3.43%
p = 4.667 MPa: exp 28.5972, calc 28.0170 mm/s -> err -2.03%
p = 5.278 MPa: exp 31.1699, calc 31.0345 mm/s -> err -0.43%
p = 5.889 MPa: exp 33.6545, calc 34.0756 mm/s -> err +1.25%
p = 6.5 MPa: exp 36.0627, calc 37.1337 mm/s -> err +2.97%
Per-fuel RMS = 0.0607 (6.07%)

Composition objective (mean per-fuel RMS) = 0.0530 (5.30%)

Bas_1 - Burn-Rate Errors

Bas_1

p = 1 MPa: exp 14.2647, calc 15.1913 mm/s -> err +6.50%
p = 1.611 MPa: exp 19.9183, calc 20.2049 mm/s -> err +1.44%
p = 2.222 MPa: exp 24.9468, calc 24.7919 mm/s -> err -0.62%
p = 2.833 MPa: exp 29.5714, calc 29.1279 mm/s -> err -1.50%
p = 3.444 MPa: exp 33.9037, calc 33.2874 mm/s -> err -1.82%
p = 4.056 MPa: exp 38.0101, calc 37.3104 mm/s -> err -1.84%
p = 4.667 MPa: exp 41.9342, calc 41.2220 mm/s -> err -1.70%
p = 5.278 MPa: exp 45.7067, calc 45.0393 mm/s -> err -1.46%
p = 5.889 MPa: exp 49.3500, calc 48.7748 mm/s -> err -1.17%
p = 6.5 MPa: exp 52.8815, calc 52.4380 mm/s -> err -0.84%

Per-fuel RMS = 0.0246 (2.46%)
Composition objective (mean per-fuel RMS) = 0.0246 (2.46%)

Bas_0 - Burn-Rate Errors

Bas_0

p = 1 MPa: exp 5.9659, calc 6.0034 mm/s -> err +0.63%
p = 1.611 MPa: exp 7.9046, calc 7.8805 mm/s -> err -0.30%
p = 2.222 MPa: exp 9.5561, calc 9.5086 mm/s -> err -0.50%
p = 2.833 MPa: exp 11.0289, calc 10.9808 mm/s -> err -0.44%
p = 3.444 MPa: exp 12.3759, calc 12.3412 mm/s -> err -0.28%
p = 4.056 MPa: exp 13.6278, calc 13.6152 mm/s -> err -0.09%
p = 4.667 MPa: exp 14.8044, calc 14.8193 mm/s -> err +0.10%
p = 5.278 MPa: exp 15.9192, calc 15.9653 mm/s -> err +0.29%
p = 5.889 MPa: exp 16.9822, calc 17.0618 mm/s -> err +0.47%
p = 6.5 MPa: exp 18.0009, calc 18.1152 mm/s -> err +0.64%
Per-fuel RMS = 0.0042 (0.42%)
Composition objective (mean per-fuel RMS) = 0.0042 (0.42%)

Overall objective (mean of the three compositions) = 0.0273 (2.73%)

Vieille Power-Law Fit (U = A*p^v, U in m/s, p in Pa)

*Experimental A/v are the supplied Vieille coefficients (the experimental burn rate is defined as exactly A*p^v, so they are exact, not re-fitted). Calculated A/v are an ordinary-least-squares fit of the model burn rates in log-log space; R2 shows how power-law-like the model curve is, and dv = v_calc - v_exp is the model's error in the pressure exponent. Numbers below reflect this report's input vector.*

Bas_2 (includes Bas_3, Bas_4) - Vieille Coefficients

Bas_2

exp: A = 1.2853E-005, v = 0.5200
calc: A = 2.7444E-006, v = 0.6182 (R2 = 0.9990); dv = +0.0982

Bas_3

exp: A = 2.2030E-006, v = 0.6400
calc: A = 2.0965E-006, v = 0.6437 (R2 = 0.9991); dv = +0.0037

Bas_4

exp: A = 6.1379E-007, v = 0.7000
calc: A = 1.3572E-006, v = 0.6473 (R2 = 0.9832); dv = -0.0527

Bas_1 - Vieille Coefficients

Bas_1

exp: A = 9.0004E-007, v = 0.7000
calc: A = 1.4864E-006, v = 0.6663 (R2 = 0.9986); dv = -0.0337

Bas_0 - Vieille Coefficients

Bas_0

exp: A = 1.7206E-006, v = 0.5900
calc: A = 1.6767E-006, v = 0.5918 (R2 = 0.9999); dv = +0.0018

Flame Structure & Heat-Flux Decomposition

q = total heat feedback to the surface (MW/m2). It splits exactly into three competing-flame pathways whose shares sum to 100%: out-kin = out-of-skeleton kinetic flame; skeleton = surface-weighted (metal burning + skeleton kinetic flame in the pores); diff = diffusion flame. h = flame heights (um). Ts_p / Ts_i = pocket / inter-pocket surface temperature (the two-temperature surface). Mean shares are the pressure-average per fuel. Numbers below reflect this report's input vector.

Bas_2 (includes Bas_3, Bas_4) - Flame Structure

Bas_2

p 1 MPa: q = 0.79 MW/m2; out-kin 64.2%, skeleton 11.7%, diff 24.1%
h[um] skel-kin 584.2, out-kin 199.8, diff 1578.4; Ts_p 612 K, Ts_i 620 K
p 1.611 MPa: q = 1.06 MW/m2; out-kin 68.5%, skeleton 17.1%, diff 14.4%

h[um] skel-kin 251.3, out-kin 144.0, diff 1947.6; Ts_p 616 K, Ts_i 631 K
p 2.222 MPa: q = 1.33 MW/m2; out-kin 68.8%, skeleton 21.4%, diff 9.8%
h[um] skel-kin 145.1, out-kin 117.7, diff 2289.2; Ts_p 619 K, Ts_i 638 K
p 2.833 MPa: q = 1.60 MW/m2; out-kin 67.8%, skeleton 25.1%, diff 7.1%
h[um] skel-kin 96.7, out-kin 102.0, diff 2610.3; Ts_p 622 K, Ts_i 643 K
p 3.444 MPa: q = 1.86 MW/m2; out-kin 66.4%, skeleton 28.1%, diff 5.5%
h[um] skel-kin 70.1, out-kin 91.4, diff 2915.3; Ts_p 624 K, Ts_i 648 K
p 4.056 MPa: q = 2.12 MW/m2; out-kin 64.9%, skeleton 30.7%, diff 4.4%
h[um] skel-kin 53.7, out-kin 83.6, diff 3207.2; Ts_p 626 K, Ts_i 652 K
p 4.667 MPa: q = 2.37 MW/m2; out-kin 63.4%, skeleton 33.0%, diff 3.6%
h[um] skel-kin 42.8, out-kin 77.5, diff 3488.1; Ts_p 627 K, Ts_i 656 K
p 5.278 MPa: q = 2.63 MW/m2; out-kin 62.0%, skeleton 35.0%, diff 3.0%
h[um] skel-kin 35.1, out-kin 72.7, diff 3759.6; Ts_p 629 K, Ts_i 659 K
p 5.889 MPa: q = 2.88 MW/m2; out-kin 60.7%, skeleton 36.7%, diff 2.6%
h[um] skel-kin 29.5, out-kin 68.7, diff 4023.0; Ts_p 630 K, Ts_i 661 K
p 6.5 MPa: q = 3.12 MW/m2; out-kin 59.4%, skeleton 38.3%, diff 2.2%
h[um] skel-kin 25.2, out-kin 65.4, diff 4279.0; Ts_p 631 K, Ts_i 664 K
mean shares: out-kin 64.6%, skeleton 27.7%, diffusion 7.7%

Bas_3

p 1 MPa: q = 1.48 MW/m2; out-kin 25.5%, skeleton 13.3%, diff 61.2%
h[um] skel-kin 369.9, out-kin 309.2, diff 337.9; Ts_p 620 K, Ts_i 620 K
p 1.611 MPa: q = 1.79 MW/m2; out-kin 33.1%, skeleton 22.9%, diff 43.9%
h[um] skel-kin 148.9, out-kin 207.8, diff 388.3; Ts_p 623 K, Ts_i 631 K
p 2.222 MPa: q = 2.12 MW/m2; out-kin 36.7%, skeleton 30.7%, diff 32.6%
h[um] skel-kin 83.3, out-kin 164.1, diff 440.6; Ts_p 626 K, Ts_i 638 K
p 2.833 MPa: q = 2.47 MW/m2; out-kin 38.2%, skeleton 36.8%, diff 25.0%
h[um] skel-kin 54.6, out-kin 139.5, diff 492.6; Ts_p 628 K, Ts_i 643 K
p 3.444 MPa: q = 2.82 MW/m2; out-kin 38.6%, skeleton 41.5%, diff 19.9%
h[um] skel-kin 39.2, out-kin 123.4, diff 543.6; Ts_p 630 K, Ts_i 648 K
p 4.056 MPa: q = 3.17 MW/m2; out-kin 38.5%, skeleton 45.3%, diff 16.2%
h[um] skel-kin 29.8, out-kin 112.0, diff 593.3; Ts_p 632 K, Ts_i 652 K
p 4.667 MPa: q = 3.51 MW/m2; out-kin 38.2%, skeleton 48.3%, diff 13.5%
h[um] skel-kin 23.7, out-kin 103.4, diff 641.7; Ts_p 633 K, Ts_i 656 K
p 5.278 MPa: q = 3.86 MW/m2; out-kin 37.7%, skeleton 50.9%, diff 11.4%
h[um] skel-kin 19.4, out-kin 96.6, diff 688.8; Ts_p 635 K, Ts_i 659 K
p 5.889 MPa: q = 4.20 MW/m2; out-kin 37.1%, skeleton 53.0%, diff 9.9%
h[um] skel-kin 16.2, out-kin 91.0, diff 734.7; Ts_p 636 K, Ts_i 661 K
p 6.5 MPa: q = 4.54 MW/m2; out-kin 36.5%, skeleton 54.9%, diff 8.6%
h[um] skel-kin 13.9, out-kin 86.3, diff 779.4; Ts_p 637 K, Ts_i 664 K
mean shares: out-kin 36.0%, skeleton 39.8%, diffusion 24.2%

Bas_4

p 1 MPa: q = 0.30 MW/m2; out-kin 5.0%, skeleton 29.5%, diff 65.5%
h[um] skel-kin 615.5, out-kin 1464.4, diff 1499.7; Ts_p 600 K, Ts_i 620 K
p 1.611 MPa: q = 0.37 MW/m2; out-kin 6.1%, skeleton 47.0%, diff 46.9%
h[um] skel-kin 244.4, out-kin 968.6, diff 1707.0; Ts_p 602 K, Ts_i 631 K
p 2.222 MPa: q = 0.46 MW/m2; out-kin 6.1%, skeleton 61.3%, diff 32.6%
h[um] skel-kin 138.1, out-kin 771.2, diff 1967.3; Ts_p 605 K, Ts_i 638 K
p 2.833 MPa: q = 0.57 MW/m2; out-kin 5.7%, skeleton 71.3%, diff 23.0%
h[um] skel-kin 92.5, out-kin 668.2, diff 2259.2; Ts_p 608 K, Ts_i 643 K
p 3.444 MPa: q = 0.69 MW/m2; out-kin 5.2%, skeleton 78.1%, diff 16.7%
h[um] skel-kin 68.1, out-kin 605.2, diff 2569.6; Ts_p 610 K, Ts_i 648 K
p 4.056 MPa: q = 0.82 MW/m2; out-kin 4.6%, skeleton 82.8%, diff 12.5%
h[um] skel-kin 53.2, out-kin 562.5, diff 2891.2; Ts_p 612 K, Ts_i 652 K
p 4.667 MPa: q = 0.96 MW/m2; out-kin 4.2%, skeleton 86.2%, diff 9.6%
h[um] skel-kin 43.4, out-kin 531.2, diff 3219.9; Ts_p 614 K, Ts_i 656 K
p 5.278 MPa: q = 1.10 MW/m2; out-kin 3.8%, skeleton 88.6%, diff 7.6%
h[um] skel-kin 36.3, out-kin 507.2, diff 3553.1; Ts_p 616 K, Ts_i 659 K
p 5.889 MPa: q = 1.25 MW/m2; out-kin 3.5%, skeleton 90.4%, diff 6.1%
h[um] skel-kin 31.1, out-kin 487.9, diff 3889.3; Ts_p 618 K, Ts_i 661 K

p 6.5 MPa: q = 1.41 MW/m2; out-kin 3.2%, skeleton 91.8%, diff 5.0%
h[um] skel-kin 27.2, out-kin 472.0, diff 4227.6; Ts_p 620 K, Ts_i 664 K
mean shares: out-kin 4.7%, skeleton 72.7%, diffusion 22.6%

Bas_1 - Flame Structure

Bas_1

p 1 MPa: q = 1.07 MW/m2; out-kin 43.4%, skeleton 40.1%, diff 16.5%
h[um] skel-kin 422.6, out-kin 216.6, diff 1728.4; Ts_p 622 K, Ts_i 628 K
p 1.611 MPa: q = 1.44 MW/m2; out-kin 46.0%, skeleton 44.2%, diff 9.8%
h[um] skel-kin 175.3, out-kin 157.6, diff 2153.9; Ts_p 626 K, Ts_i 638 K
p 2.222 MPa: q = 1.81 MW/m2; out-kin 45.6%, skeleton 47.9%, diff 6.6%
h[um] skel-kin 98.7, out-kin 130.2, diff 2557.4; Ts_p 630 K, Ts_i 646 K
p 2.833 MPa: q = 2.18 MW/m2; out-kin 44.3%, skeleton 51.0%, diff 4.7%
h[um] skel-kin 64.5, out-kin 113.9, diff 2944.1; Ts_p 632 K, Ts_i 651 K
p 3.444 MPa: q = 2.55 MW/m2; out-kin 42.8%, skeleton 53.6%, diff 3.6%
h[um] skel-kin 46.1, out-kin 102.8, diff 3317.2; Ts_p 635 K, Ts_i 656 K
p 4.056 MPa: q = 2.92 MW/m2; out-kin 41.3%, skeleton 55.9%, diff 2.8%
h[um] skel-kin 34.9, out-kin 94.8, diff 3679.0; Ts_p 637 K, Ts_i 660 K
p 4.667 MPa: q = 3.29 MW/m2; out-kin 39.8%, skeleton 57.9%, diff 2.3%
h[um] skel-kin 27.5, out-kin 88.6, diff 4031.2; Ts_p 639 K, Ts_i 664 K
p 5.278 MPa: q = 3.66 MW/m2; out-kin 38.5%, skeleton 59.6%, diff 1.9%
h[um] skel-kin 22.4, out-kin 83.7, diff 4375.0; Ts_p 640 K, Ts_i 667 K
p 5.889 MPa: q = 4.02 MW/m2; out-kin 37.2%, skeleton 61.2%, diff 1.6%
h[um] skel-kin 18.6, out-kin 79.6, diff 4711.4; Ts_p 642 K, Ts_i 669 K
p 6.5 MPa: q = 4.39 MW/m2; out-kin 36.1%, skeleton 62.6%, diff 1.4%
h[um] skel-kin 15.8, out-kin 76.2, diff 5041.2; Ts_p 643 K, Ts_i 672 K
mean shares: out-kin 41.5%, skeleton 53.4%, diffusion 5.1%

Bas_0 - Flame Structure

Bas_0

p 1 MPa: q = 1.83 MW/m2; out-kin 73.7%, skeleton 1.5%, diff 24.8%
h[um] skel-kin 1887.2, out-kin 65.7, diff 626.1; Ts_p 818 K, Ts_i 836 K
p 1.611 MPa: q = 2.33 MW/m2; out-kin 83.0%, skeleton 1.4%, diff 15.7%
h[um] skel-kin 1581.8, out-kin 47.7, diff 778.7; Ts_p 824 K, Ts_i 852 K
p 2.222 MPa: q = 2.78 MW/m2; out-kin 87.6%, skeleton 1.3%, diff 11.1%
h[um] skel-kin 1419.4, out-kin 39.0, diff 915.0; Ts_p 828 K, Ts_i 862 K
p 2.833 MPa: q = 3.20 MW/m2; out-kin 90.3%, skeleton 1.2%, diff 8.5%
h[um] skel-kin 1313.6, out-kin 33.7, diff 1039.7; Ts_p 831 K, Ts_i 871 K
p 3.444 MPa: q = 3.60 MW/m2; out-kin 92.1%, skeleton 1.1%, diff 6.8%
h[um] skel-kin 1237.0, out-kin 30.0, diff 1155.6; Ts_p 834 K, Ts_i 877 K
p 4.056 MPa: q = 3.97 MW/m2; out-kin 93.3%, skeleton 1.1%, diff 5.6%
h[um] skel-kin 1177.7, out-kin 27.3, diff 1264.5; Ts_p 836 K, Ts_i 883 K
p 4.667 MPa: q = 4.33 MW/m2; out-kin 94.2%, skeleton 1.0%, diff 4.8%
h[um] skel-kin 1129.9, out-kin 25.2, diff 1367.6; Ts_p 838 K, Ts_i 888 K
p 5.278 MPa: q = 4.67 MW/m2; out-kin 94.9%, skeleton 1.0%, diff 4.1%
h[um] skel-kin 1090.1, out-kin 23.5, diff 1465.8; Ts_p 840 K, Ts_i 892 K
p 5.889 MPa: q = 5.00 MW/m2; out-kin 95.4%, skeleton 0.9%, diff 3.6%
h[um] skel-kin 1056.2, out-kin 22.1, diff 1560.0; Ts_p 841 K, Ts_i 896 K
p 6.5 MPa: q = 5.32 MW/m2; out-kin 95.9%, skeleton 0.9%, diff 3.2%
h[um] skel-kin 1026.7, out-kin 20.9, diff 1650.4; Ts_p 843 K, Ts_i 900 K
mean shares: out-kin 90.0%, skeleton 1.1%, diffusion 8.8%

Propellant Composition "Bas_2"

At pressure 1 MPa

Experimental burning rate 16.94 mm/s
Calculated burning rate 14.36 mm/s
First Conditional Fuel

Surface temperature 620.36 K

Linear burning rate 11.66 mm/s
Mass decomposition rate 19.35 $kg\cdot s^{-1}\cdot m^{-2}$
Kinetic flame heat flux density to surface 1.48e+06 W/m^2
Kinetic flame height 209.07 μm
Average kinetic flame density 1.83 kg/m^3
Average kinetic flame temperature 2,163.68 K
Binder sublimation heat flux density 1.48e+06 W/m^2
Surface heat balance error -4.37e-04 W/m^2

Secondary Conditional Fuel

Surface temperature 611.74 K
Linear burning rate 7.47 mm/s
Mass decomposition rate 12.4 $kg\cdot s^{-1}\cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 877,543.79 W/m^2
Kinetic flame height 199.79 μm
Average kinetic flame density 2.67 kg/m^3
Average kinetic flame temperature 1,488.37 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 167,628.46 W/m^2
Kinetic flame height 584.18 μm
Average kinetic flame density 3.6 kg/m^3
Average kinetic flame temperature 1,101.37 K
Heat flux density due to metal combustion, 48,498.9 W/m^2
Effective thermal conductivity 0.4 $W/(m\cdot K)$
Conductive thermal conductivity 0.4 $W/(m\cdot K)$
Radiative thermal conductivity 2.28e-04 $W/(m\cdot K)$
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 3.51e-05 m
Pore diameter 1.94e-06 m

Diffusion flame heat flux density 189,579.33 W/m^2
Diffusion flame height 1,578.37 μm
Heat flux density to surface within skeletal structure 91,970.79 W/m^2
Heat flux density to surface outside skeletal structure 504,113.99 W/m^2
Total heat flux density to surface 785,664.11 W/m^2
Binder sublimation heat flux density 785,664.12 W/m^2
Surface heat balance error -3.85e-04 W/m^2

At pressure 1.61 MPa

Experimental burning rate 21.71 mm/s
Calculated burning rate 18.78 mm/s

First Conditional Fuel

Surface temperature 630.61 K
Linear burning rate 19.49 mm/s
Mass decomposition rate 32.33 $kg\cdot s^{-1}\cdot m^{-2}$
Kinetic flame heat flux density to surface 2.96e+06 W/m^2
Kinetic flame height 103.8 μm
Average kinetic flame density 2.95 kg/m^3
Average kinetic flame temperature 2,168.81 K
Binder sublimation heat flux density 2.96e+06 W/m^2
Surface heat balance error -9.88e-04 W/m^2

Secondary Conditional Fuel

Surface temperature 615.78 K
Linear burning rate 9.22 mm/s
Mass decomposition rate 15.3 $kg\cdot s^{-1}\cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.21e+06 W/m^2
Kinetic flame height 143.98 μm
Average kinetic flame density 4.29 kg/m^3
Average kinetic flame temperature 1,490.39 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 388,002.6 W/m²

Kinetic flame height 251.34 μm

Average kinetic flame density 5.8 kg/m³

Average kinetic flame temperature 1,103.39 K

Heat flux density due to metal combustion, 63,908.82 W/m²

Effective thermal conductivity 0.4 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 2.28e-04 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 2.66e-05 m

Pore diameter 1.91e-06 m

Diffusion flame heat flux density 153,433.03 W/m²

Diffusion flame height 1,947.57 μm

Heat flux density to surface within skeletal structure 181,318.62 W/m²

Heat flux density to surface outside skeletal structure 727,456.55 W/m²

Total heat flux density to surface 1.06e+06 W/m²

Binder sublimation heat flux density 1.06e+06 W/m²

Surface heat balance error -2.48e-04 W/m²

<u>*At pressure 2.22 MPa*</u>

Experimental burning rate 25.67 mm/s

Calculated burning rate 22.73 mm/s

First Conditional Fuel

Surface temperature 637.85 K

Linear burning rate 27.72 mm/s

Mass decomposition rate 45.98 $kg \cdot s^{-1} \cdot m^{-2}$

Kinetic flame heat flux density to surface 4.71e+06 W/m²

Kinetic flame height 65.1 μm

Average kinetic flame density 4.06 kg/m³

Average kinetic flame temperature 2,172.42 K

Binder sublimation heat flux density 4.71e+06 W/m²

Surface heat balance error -0.0016 W/m²

Secondary Conditional Fuel

Surface temperature 618.93 K

Linear burning rate 10.84 mm/s

Mass decomposition rate 17.98 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.48e+06 W/m²

Kinetic flame height 117.71 μm

Average kinetic flame density 5.91 kg/m³

Average kinetic flame temperature 1,491.96 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 669,934.27 W/m²

Kinetic flame height 145.1 μm

Average kinetic flame density 7.98 kg/m³

Average kinetic flame temperature 1,104.96 K

Heat flux density due to metal combustion, 79,011.49 W/m²

Effective thermal conductivity 0.4 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 2.28e-04 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 2.15e-05 m

Pore diameter 1.88e-06 m

Diffusion flame heat flux density 130,397.51 W/m²

Diffusion flame height 2,289.21 μm

Heat flux density to surface within skeletal structure 285,909.55 W/m²

Heat flux density to surface outside skeletal structure 917,060.12 W/m²

Total heat flux density to surface 1.33e+06 W/m²

Binder sublimation heat flux density 1.33e+06 W/m²
Surface heat balance error 5.4e-04 W/m²

At pressure 2.83 MPa
Experimental burning rate 29.12 mm/s
Calculated burning rate 26.41 mm/s
First Conditional Fuel
Surface temperature 643.49 K
Linear burning rate 36.27 mm/s
Mass decomposition rate 60.17 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 6.68e+06 W/m²
Kinetic flame height 45.88 μm
Average kinetic flame density 5.17 kg/m³
Average kinetic flame temperature 2,175.24 K
Binder sublimation heat flux density 6.68e+06 W/m²
Surface heat balance error 5.65e-04 W/m²
Secondary Conditional Fuel
Surface temperature 621.51 K
Linear burning rate 12.36 mm/s
Mass decomposition rate 20.5 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 1.71e+06 W/m²
Kinetic flame height 102.01 μm
Average kinetic flame density 7.53 kg/m³
Average kinetic flame temperature 1,493.25 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1e+06 W/m²
Kinetic flame height 96.66 μm
Average kinetic flame density 10.17 kg/m³
Average kinetic flame temperature 1,106.25 K
Heat flux density due to metal combustion, 93,864.73 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 2.28e-04 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.81e-05 m
Pore diameter 1.87e-06 m
Diffusion flame heat flux density 114,258.94 W/m²
Diffusion flame height 2,610.29 μm
Heat flux density to surface within skeletal structure 400,768.29 W/m²
Heat flux density to surface outside skeletal structure 1.08e+06 W/m²
Total heat flux density to surface 1.6e+06 W/m²
Binder sublimation heat flux density 1.6e+06 W/m²
Surface heat balance error -0.0025 W/m²

At pressure 3.44 MPa
Experimental burning rate 32.23 mm/s
Calculated burning rate 29.88 mm/s
First Conditional Fuel
Surface temperature 648.12 K
Linear burning rate 45.09 mm/s
Mass decomposition rate 74.81 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 8.82e+06 W/m²
Kinetic flame height 34.67 μm
Average kinetic flame density 6.28 kg/m³
Average kinetic flame temperature 2,177.56 K
Binder sublimation heat flux density 8.82e+06 W/m²
Surface heat balance error -0.0035 W/m²
Secondary Conditional Fuel

Surface temperature 623.69 K
Linear burning rate 13.8 mm/s
Mass decomposition rate 22.9 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 1.91e+06 W/m^2
Kinetic flame height 91.36 μm
Average kinetic flame density 9.15 kg/m^3
Average kinetic flame temperature 1,494.35 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1.38e+06 W/m^2
Kinetic flame height 70.07 μm
Average kinetic flame density 12.35 kg/m^3
Average kinetic flame temperature 1,107.35 K
Heat flux density due to metal combustion, 108,513.21 W/m^2
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 2.29e-04 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.56e-05 m
Pore diameter 1.85e-06 m
Diffusion flame heat flux density 102,230.23 W/m^2
Diffusion flame height 2,915.29 μm
Heat flux density to surface within skeletal structure 523,009.33 W/m^2
Heat flux density to surface outside skeletal structure 1.24e+06 W/m^2
Total heat flux density to surface 1.86e+06 W/m^2
Binder sublimation heat flux density 1.86e+06 W/m^2
Surface heat balance error -7.56e-04 W/m^2

At pressure 4.06 MPa
Experimental burning rate 35.09 mm/s
Calculated burning rate 33.2 mm/s
First Conditional Fuel
Surface temperature 652.07 K
Linear burning rate 54.15 mm/s
Mass decomposition rate 89.83 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.11e+07 W/m^2
Kinetic flame height 27.46 μm
Average kinetic flame density 7.39 kg/m^3
Average kinetic flame temperature 2,179.54 K
Binder sublimation heat flux density 1.11e+07 W/m^2
Surface heat balance error 0.0033 W/m^2
Secondary Conditional Fuel
Surface temperature 625.59 K
Linear burning rate 15.19 mm/s
Mass decomposition rate 25.19 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 2.08e+06 W/m^2
Kinetic flame height 83.56 μm
Average kinetic flame density 10.76 kg/m^3
Average kinetic flame temperature 1,495.3 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1.8e+06 W/m^2
Kinetic flame height 53.7 μm
Average kinetic flame density 14.52 kg/m^3
Average kinetic flame temperature 1,108.3 K
Heat flux density due to metal combustion, 122,988.48 W/m^2
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 2.29e-04 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.37e-05 m
Pore diameter 1.84e-06 m
Diffusion flame heat flux density 92,866.51 W/m²
Diffusion flame height 3,207.19 μm
Heat flux density to surface within skeletal structure 650,784.12 W/m²
Heat flux density to surface outside skeletal structure 1.38e+06 W/m²
Total heat flux density to surface 2.12e+06 W/m²
Binder sublimation heat flux density 2.12e+06 W/m²
Surface heat balance error -5.43e-04 W/m²

At pressure 4.67 MPa
Experimental burning rate 37.75 mm/s
Calculated burning rate 36.39 mm/s
First Conditional Fuel
Surface temperature 655.52 K
Linear burning rate 63.41 mm/s
Mass decomposition rate 105.2 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.36e+07 W/m²
Kinetic flame height 22.48 μm
Average kinetic flame density 8.49 kg/m³
Average kinetic flame temperature 2,181.26 K
Binder sublimation heat flux density 1.36e+07 W/m²
Surface heat balance error 0.003 W/m²
Secondary Conditional Fuel
Surface temperature 627.28 K
Linear burning rate 16.52 mm/s
Mass decomposition rate 27.4 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 2.24e+06 W/m²
Kinetic flame height 77.54 μm
Average kinetic flame density 12.38 kg/m³
Average kinetic flame temperature 1,496.14 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 2.25e+06 W/m²
Kinetic flame height 42.8 μm
Average kinetic flame density 16.7 kg/m³
Average kinetic flame temperature 1,109.14 K
Heat flux density due to metal combustion, 137,313.34 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 2.29e-04 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.23e-05 m
Pore diameter 1.82e-06 m
Diffusion flame heat flux density 85,338.99 W/m²
Diffusion flame height 3,488.12 μm
Heat flux density to surface within skeletal structure 782,828.9 W/m²
Heat flux density to surface outside skeletal structure 1.51e+06 W/m²
Total heat flux density to surface 2.37e+06 W/m²
Binder sublimation heat flux density 2.37e+06 W/m²
Surface heat balance error 1.19e-04 W/m²

At pressure 5.28 MPa
Experimental burning rate 40.24 mm/s
Calculated burning rate 39.46 mm/s
First Conditional Fuel
Surface temperature 658.58 K
Linear burning rate 72.87 mm/s

Mass decomposition rate 120.89 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.62e+07 W/m^2
Kinetic flame height 18.87 μm
Average kinetic flame density 9.6 kg/m^3
Average kinetic flame temperature 2,182.79 K
Binder sublimation heat flux density 1.62e+07 W/m^2
Surface heat balance error -0.0041 W/m^2

Secondary Conditional Fuel

Surface temperature 628.78 K
Linear burning rate 17.8 mm/s
Mass decomposition rate 29.53 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 2.39e+06 W/m^2
Kinetic flame height 72.72 μm
Average kinetic flame density 13.99 kg/m^3
Average kinetic flame temperature 1,496.89 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 2.74e+06 W/m^2
Kinetic flame height 35.13 μm
Average kinetic flame density 18.87 kg/m^3
Average kinetic flame temperature 1,109.89 K
Heat flux density due to metal combustion, 151,504.8 W/m^2
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 2.29e-04 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.11e-05 m
Pore diameter 1.81e-06 m

Diffusion flame heat flux density 79,135.68 W/m^2
Diffusion flame height 3,759.64 μm
Heat flux density to surface within skeletal structure 918,238.84 W/m^2
Heat flux density to surface outside skeletal structure 1.63e+06 W/m^2
Total heat flux density to surface 2.63e+06 W/m^2
Binder sublimation heat flux density 2.63e+06 W/m^2
Surface heat balance error -0.0011 W/m^2

<u>*At pressure 5.89 MPa*</u>

Experimental burning rate 42.6 mm/s
Calculated burning rate 42.45 mm/s

First Conditional Fuel

Surface temperature 661.34 K
Linear burning rate 82.49 mm/s
Mass decomposition rate 136.85 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.89e+07 W/m^2
Kinetic flame height 16.15 μm
Average kinetic flame density 10.7 kg/m^3
Average kinetic flame temperature 2,184.17 K
Binder sublimation heat flux density 1.89e+07 W/m^2
Surface heat balance error 2.82e-04 W/m^2

Secondary Conditional Fuel

Surface temperature 630.15 K
Linear burning rate 19.05 mm/s
Mass decomposition rate 31.6 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 2.52e+06 W/m^2
Kinetic flame height 68.74 μm
Average kinetic flame density 15.61 kg/m^3
Average kinetic flame temperature 1,497.58 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 3.26e+06 W/m²
Kinetic flame height 29.49 µm
Average kinetic flame density 21.05 kg/m³
Average kinetic flame temperature 1,110.58 K
Heat flux density due to metal combustion, 165,576.03 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 2.29e-04 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.02e-05 m
Pore diameter 1.8e-06 m
Diffusion flame heat flux density 73,921.95 W/m²
Diffusion flame height 4,022.95 µm
Heat flux density to surface within skeletal structure 1.06e+06 W/m²
Heat flux density to surface outside skeletal structure 1.75e+06 W/m²
Total heat flux density to surface 2.88e+06 W/m²
Binder sublimation heat flux density 2.88e+06 W/m²
Surface heat balance error -1.75e-04 W/m²

At pressure 6.5 MPa
Experimental burning rate 44.85 mm/s
Calculated burning rate 45.35 mm/s
First Conditional Fuel
Surface temperature 663.85 K
Linear burning rate 92.27 mm/s
Mass decomposition rate 153.08 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 2.17e+07 W/m²
Kinetic flame height 14.05 µm
Average kinetic flame density 11.8 kg/m³
Average kinetic flame temperature 2,185.43 K
Binder sublimation heat flux density 2.17e+07 W/m²
Surface heat balance error 0.0092 W/m²
Secondary Conditional Fuel
Surface temperature 631.41 K
Linear burning rate 20.26 mm/s
Mass decomposition rate 33.61 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 2.65e+06 W/m²
Kinetic flame height 65.38 µm
Average kinetic flame density 17.22 kg/m³
Average kinetic flame temperature 1,498.2 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 3.81e+06 W/m²
Kinetic flame height 25.21 µm
Average kinetic flame density 23.22 kg/m³
Average kinetic flame temperature 1,111.2 K
Heat flux density due to metal combustion, 179,537.53 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 2.29e-04 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 9.38e-06 m
Pore diameter 1.8e-06 m
Diffusion flame heat flux density 69,469.11 W/m²
Diffusion flame height 4,279.01 µm
Heat flux density to surface within skeletal structure 1.2e+06 W/m²
Heat flux density to surface outside skeletal structure 1.86e+06 W/m²
Total heat flux density to surface 3.12e+06 W/m²
Binder sublimation heat flux density 3.12e+06 W/m²

Surface heat balance error 0.0012 W/m²

Propellant Composition "Bas_3"

At pressure 1 MPa
Experimental burning rate 15.24 mm/s
Calculated burning rate 15.57 mm/s
First Conditional Fuel
Surface temperature 620.36 K
Linear burning rate 11.66 mm/s
Mass decomposition rate 19.35 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.48e+06 W/m²
Kinetic flame height 209.07 μm
Average kinetic flame density 1.83 kg/m³
Average kinetic flame temperature 2,163.68 K
Binder sublimation heat flux density 1.48e+06 W/m²
Surface heat balance error -4.37e-04 W/m²
Secondary Conditional Fuel
Surface temperature 620.36 K
Linear burning rate 11.66 mm/s
Mass decomposition rate 19.35 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 564,295.59 W/m²
Kinetic flame height 309.17 μm
Average kinetic flame density 2.66 kg/m³
Average kinetic flame temperature 1,492.68 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 541,370.67 W/m²
Kinetic flame height 369.92 μm
Average kinetic flame density 2.45 kg/m³
Average kinetic flame temperature 1,621.68 K
Heat flux density due to metal combustion, 47,851.91 W/m²
Effective thermal conductivity 0.22 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 3.99e-04 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 1.95e-05 m
Pore diameter 1.87e-06 m
Diffusion flame heat flux density 903,760.77 W/m²
Diffusion flame height 337.88 μm
Heat flux density to surface within skeletal structure 196,003.11 W/m²
Heat flux density to surface outside skeletal structure 376,584.37 W/m²
Total heat flux density to surface 1.48e+06 W/m²
Binder sublimation heat flux density 1.48e+06 W/m²
Surface heat balance error 2.68e-04 W/m²

At pressure 1.61 MPa
Experimental burning rate 20.68 mm/s
Calculated burning rate 20.67 mm/s
First Conditional Fuel
Surface temperature 630.61 K
Linear burning rate 19.49 mm/s
Mass decomposition rate 32.33 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 2.96e+06 W/m²
Kinetic flame height 103.8 μm
Average kinetic flame density 2.95 kg/m³
Average kinetic flame temperature 2,168.81 K
Binder sublimation heat flux density 2.96e+06 W/m²
Surface heat balance error -9.88e-04 W/m²

Secondary Conditional Fuel

Surface temperature 623.11 K

Linear burning rate 13.4 mm/s

Mass decomposition rate 22.24 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 838,248.12 W/m^2

Kinetic flame height 207.8 μm

Average kinetic flame density 4.28 kg/m^3

Average kinetic flame temperature 1,494.06 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.34e+06 W/m^2

Kinetic flame height 148.95 μm

Average kinetic flame density 3.94 kg/m^3

Average kinetic flame temperature 1,623.06 K

Heat flux density due to metal combustion, 57,442.72 W/m^2

Effective thermal conductivity 0.22 W/(m·K)

Conductive thermal conductivity 0.22 W/(m·K)

Radiative thermal conductivity 3.99e-04 W/(m·K)

Conductive thermal conductivity balance error -5.5152085010057306E-08

Skeleton layer thickness 1.62e-05 m

Pore diameter 1.85e-06 m

Diffusion flame heat flux density 785,612.93 W/m^2

Diffusion flame height 388.35 μm

Heat flux density to surface within skeletal structure 410,226.04 W/m^2

Heat flux density to surface outside skeletal structure 592,647.91 W/m^2

Total heat flux density to surface 1.79e+06 W/m^2

Binder sublimation heat flux density 1.79e+06 W/m^2

Surface heat balance error -8.99e-05 W/m^2

At pressure 2.22 MPa

Experimental burning rate 25.41 mm/s

Calculated burning rate 25.19 mm/s

First Conditional Fuel

Surface temperature 637.85 K

Linear burning rate 27.72 mm/s

Mass decomposition rate 45.98 $kg \cdot s^{-1} \cdot m^{-2}$

Kinetic flame heat flux density to surface 4.71e+06 W/m^2

Kinetic flame height 65.1 μm

Average kinetic flame density 4.06 kg/m^3

Average kinetic flame temperature 2,172.42 K

Binder sublimation heat flux density 4.71e+06 W/m^2

Surface heat balance error -0.0016 W/m^2

Secondary Conditional Fuel

Surface temperature 625.62 K

Linear burning rate 15.21 mm/s

Mass decomposition rate 25.23 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.06e+06 W/m^2

Kinetic flame height 164.09 μm

Average kinetic flame density 5.9 kg/m^3

Average kinetic flame temperature 1,495.31 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 2.4e+06 W/m^2

Kinetic flame height 83.28 μm

Average kinetic flame density 5.43 kg/m^3

Average kinetic flame temperature 1,624.31 K

Heat flux density due to metal combustion, 67,791.41 W/m^2

Effective thermal conductivity 0.22 W/(m·K)

Conductive thermal conductivity 0.22 W/(m·K)

Radiative thermal conductivity 3.99e-04 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 1.37e-05 m
Pore diameter 1.84e-06 m
Diffusion flame heat flux density 691,875.21 W/m²
Diffusion flame height 440.6 µm
Heat flux density to surface within skeletal structure 653,017.71 W/m²
Heat flux density to surface outside skeletal structure 779,316.87 W/m²
Total heat flux density to surface 2.12e+06 W/m²
Binder sublimation heat flux density 2.12e+06 W/m²
Surface heat balance error 4.87e-05 W/m²

At pressure 2.83 MPa
Experimental burning rate 29.68 mm/s
Calculated burning rate 29.42 mm/s
First Conditional Fuel
Surface temperature 643.49 K
Linear burning rate 36.27 mm/s
Mass decomposition rate 60.17 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 6.68e+06 W/m²
Kinetic flame height 45.88 µm
Average kinetic flame density 5.17 kg/m³
Average kinetic flame temperature 2,175.24 K
Binder sublimation heat flux density 6.68e+06 W/m²
Surface heat balance error 5.65e-04 W/m²
Secondary Conditional Fuel
Surface temperature 627.86 K
Linear burning rate 17 mm/s
Mass decomposition rate 28.21 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 1.25e+06 W/m²
Kinetic flame height 139.47 µm
Average kinetic flame density 7.51 kg/m³
Average kinetic flame temperature 1,496.43 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 3.66e+06 W/m²
Kinetic flame height 54.55 µm
Average kinetic flame density 6.92 kg/m³
Average kinetic flame temperature 1,625.43 K
Heat flux density due to metal combustion, 78,478.26 W/m²
Effective thermal conductivity 0.22 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 3.99e-04 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 1.18e-05 m
Pore diameter 1.82e-06 m
Diffusion flame heat flux density 618,381.64 W/m²
Diffusion flame height 492.6 µm
Heat flux density to surface within skeletal structure 908,648.71 W/m²
Heat flux density to surface outside skeletal structure 942,563.51 W/m²
Total heat flux density to surface 2.47e+06 W/m²
Binder sublimation heat flux density 2.47e+06 W/m²
Surface heat balance error 0.001 W/m²

At pressure 3.44 MPa
Experimental burning rate 33.63 mm/s
Calculated burning rate 33.45 mm/s
First Conditional Fuel
Surface temperature 648.12 K

Linear burning rate 45.09 mm/s
Mass decomposition rate 74.81 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 8.82e+06 W/m^2
Kinetic flame height 34.67 μm
Average kinetic flame density 6.28 kg/m^3
Average kinetic flame temperature 2,177.56 K
Binder sublimation heat flux density 8.82e+06 W/m^2
Surface heat balance error -0.0035 W/m^2

Secondary Conditional Fuel

Surface temperature 629.85 K
Linear burning rate 18.76 mm/s
Mass decomposition rate 31.13 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.41e+06 W/m^2
Kinetic flame height 123.45 μm
Average kinetic flame density 9.13 kg/m^3
Average kinetic flame temperature 1,497.42 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 5.09e+06 W/m^2
Kinetic flame height 39.16 μm
Average kinetic flame density 8.41 kg/m^3
Average kinetic flame temperature 1,626.42 K
Heat flux density due to metal combustion, 89,304.68 W/m^2
Effective thermal conductivity 0.22 $W/(m \cdot K)$
Conductive thermal conductivity 0.22 $W/(m \cdot K)$
Radiative thermal conductivity 3.99e-04 $W/(m \cdot K)$
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 1.04e-05 m
Pore diameter 1.81e-06 m

Diffusion flame heat flux density 560,009.36 W/m^2
Diffusion flame height 543.59 μm
Heat flux density to surface within skeletal structure 1.17e+06 W/m^2
Heat flux density to surface outside skeletal structure 1.09e+06 W/m^2
Total heat flux density to surface 2.82e+06 W/m^2
Binder sublimation heat flux density 2.82e+06 W/m^2
Surface heat balance error -0.0013 W/m^2

<u>*At pressure 4.06 MPa*</u>

Experimental burning rate 37.34 mm/s
Calculated burning rate 37.32 mm/s

First Conditional Fuel

Surface temperature 652.07 K
Linear burning rate 54.15 mm/s
Mass decomposition rate 89.83 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.11e+07 W/m^2
Kinetic flame height 27.46 μm
Average kinetic flame density 7.39 kg/m^3
Average kinetic flame temperature 2,179.54 K
Binder sublimation heat flux density 1.11e+07 W/m^2
Surface heat balance error 0.0033 W/m^2

Secondary Conditional Fuel

Surface temperature 631.63 K
Linear burning rate 20.48 mm/s
Mass decomposition rate 33.98 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.55e+06 W/m^2
Kinetic flame height 112.04 μm
Average kinetic flame density 10.74 kg/m^3
Average kinetic flame temperature 1,498.31 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 6.67e+06 W/m²

Kinetic flame height 29.83 μm

Average kinetic flame density 9.89 kg/m³

Average kinetic flame temperature 1,627.31 K

Heat flux density due to metal combustion, 100,170.66 W/m²

Effective thermal conductivity 0.22 W/(m·K)

Conductive thermal conductivity 0.22 W/(m·K)

Radiative thermal conductivity 3.99e-04 W/(m·K)

Conductive thermal conductivity balance error -5.5152085010057306E-08

Skeleton layer thickness 9.25e-06 m

Pore diameter 1.8e-06 m

Diffusion flame heat flux density 512,780.65 W/m²

Diffusion flame height 593.31 μm

Heat flux density to surface within skeletal structure 1.43e+06 W/m²

Heat flux density to surface outside skeletal structure 1.22e+06 W/m²

Total heat flux density to surface 3.17e+06 W/m²

Binder sublimation heat flux density 3.17e+06 W/m²

Surface heat balance error 4.78e-04 W/m²

At pressure 4.67 MPa

Experimental burning rate 40.85 mm/s

Calculated burning rate 41.06 mm/s

First Conditional Fuel

Surface temperature 655.52 K

Linear burning rate 63.41 mm/s

Mass decomposition rate 105.2 kg·s⁻¹·m⁻²

Kinetic flame heat flux density to surface 1.36e+07 W/m²

Kinetic flame height 22.48 μm

Average kinetic flame density 8.49 kg/m³

Average kinetic flame temperature 2,181.26 K

Binder sublimation heat flux density 1.36e+07 W/m²

Surface heat balance error 0.003 W/m²

Secondary Conditional Fuel

Surface temperature 633.23 K

Linear burning rate 22.15 mm/s

Mass decomposition rate 36.75 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.67e+06 W/m²

Kinetic flame height 103.4 μm

Average kinetic flame density 12.36 kg/m³

Average kinetic flame temperature 1,499.11 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 8.4e+06 W/m²

Kinetic flame height 23.69 μm

Average kinetic flame density 11.38 kg/m³

Average kinetic flame temperature 1,628.11 K

Heat flux density due to metal combustion, 111,023.52 W/m²

Effective thermal conductivity 0.22 W/(m·K)

Conductive thermal conductivity 0.22 W/(m·K)

Radiative thermal conductivity 3.99e-04 W/(m·K)

Conductive thermal conductivity balance error -5.5152085010057306E-08

Skeleton layer thickness 8.34e-06 m

Pore diameter 1.78e-06 m

Diffusion flame heat flux density 473,858.26 W/m²

Diffusion flame height 641.71 μm

Heat flux density to surface within skeletal structure 1.7e+06 W/m²

Heat flux density to surface outside skeletal structure 1.34e+06 W/m²

Total heat flux density to surface 3.51e+06 W/m²

Binder sublimation heat flux density 3.51e+06 W/m^2
Surface heat balance error -5.51e-04 W/m^2

At pressure 5.28 MPa
Experimental burning rate 44.2 mm/s
Calculated burning rate 44.69 mm/s
First Conditional Fuel
Surface temperature 658.58 K
Linear burning rate 72.87 mm/s
Mass decomposition rate 120.89 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.62e+07 W/m^2
Kinetic flame height 18.87 μm
Average kinetic flame density 9.6 kg/m^3
Average kinetic flame temperature 2,182.79 K
Binder sublimation heat flux density 1.62e+07 W/m^2
Surface heat balance error -0.0041 W/m^2
Secondary Conditional Fuel
Surface temperature 634.68 K
Linear burning rate 23.78 mm/s
Mass decomposition rate 39.45 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 1.79e+06 W/m^2
Kinetic flame height 96.57 μm
Average kinetic flame density 13.97 kg/m^3
Average kinetic flame temperature 1,499.84 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1.03e+07 W/m^2
Kinetic flame height 19.39 μm
Average kinetic flame density 12.86 kg/m^3
Average kinetic flame temperature 1,628.84 K
Heat flux density due to metal combustion, 121,834.82 W/m^2
Effective thermal conductivity 0.22 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 3.99e-04 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 7.59e-06 m
Pore diameter 1.77e-06 m
Diffusion flame heat flux density 441,241.32 W/m^2
Diffusion flame height 688.81 μm
Heat flux density to surface within skeletal structure 1.96e+06 W/m^2
Heat flux density to surface outside skeletal structure 1.45e+06 W/m^2
Total heat flux density to surface 3.86e+06 W/m^2
Binder sublimation heat flux density 3.86e+06 W/m^2
Surface heat balance error 0.0017 W/m^2

At pressure 5.89 MPa
Experimental burning rate 47.41 mm/s
Calculated burning rate 48.21 mm/s
First Conditional Fuel
Surface temperature 661.34 K
Linear burning rate 82.49 mm/s
Mass decomposition rate 136.85 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.89e+07 W/m^2
Kinetic flame height 16.15 μm
Average kinetic flame density 10.7 kg/m^3
Average kinetic flame temperature 2,184.17 K
Binder sublimation heat flux density 1.89e+07 W/m^2
Surface heat balance error 2.82e-04 W/m^2
Secondary Conditional Fuel

Surface temperature 636.01 K
Linear burning rate 25.36 mm/s
Mass decomposition rate 42.07 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 1.9e+06 W/m^2
Kinetic flame height 91 μm
Average kinetic flame density 15.58 kg/m^3
Average kinetic flame temperature 1,500.5 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1.22e+07 W/m^2
Kinetic flame height 16.25 μm
Average kinetic flame density 14.34 kg/m^3
Average kinetic flame temperature 1,629.5 K
Heat flux density due to metal combustion, 132,589.21 W/m^2
Effective thermal conductivity 0.22 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 3.99e-04 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 6.97e-06 m
Pore diameter 1.77e-06 m
Diffusion flame heat flux density 413,503.88 W/m^2
Diffusion flame height 734.69 μm
Heat flux density to surface within skeletal structure 2.23e+06 W/m^2
Heat flux density to surface outside skeletal structure 1.56e+06 W/m^2
Total heat flux density to surface 4.2e+06 W/m^2
Binder sublimation heat flux density 4.2e+06 W/m^2
Surface heat balance error -0.0016 W/m^2

At pressure 6.5 MPa
Experimental burning rate 50.5 mm/s
Calculated burning rate 51.65 mm/s
First Conditional Fuel
Surface temperature 663.85 K
Linear burning rate 92.27 mm/s
Mass decomposition rate 153.08 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 2.17e+07 W/m^2
Kinetic flame height 14.05 μm
Average kinetic flame density 11.8 kg/m^3
Average kinetic flame temperature 2,185.43 K
Binder sublimation heat flux density 2.17e+07 W/m^2
Surface heat balance error 0.0092 W/m^2
Secondary Conditional Fuel
Surface temperature 637.23 K
Linear burning rate 26.9 mm/s
Mass decomposition rate 44.63 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 2e+06 W/m^2
Kinetic flame height 86.34 μm
Average kinetic flame density 17.19 kg/m^3
Average kinetic flame temperature 1,501.12 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1.43e+07 W/m^2
Kinetic flame height 13.87 μm
Average kinetic flame density 15.83 kg/m^3
Average kinetic flame temperature 1,630.12 K
Heat flux density due to metal combustion, 143,278.68 W/m^2
Effective thermal conductivity 0.22 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 3.99e-04 W/(m·K)

Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 6.45e-06 m
Pore diameter 1.76e-06 m
Diffusion flame heat flux density 389,611.96 W/m²
Diffusion flame height 779.43 µm
Heat flux density to surface within skeletal structure 2.49e+06 W/m²
Heat flux density to surface outside skeletal structure 1.66e+06 W/m²
Total heat flux density to surface 4.54e+06 W/m²
Binder sublimation heat flux density 4.54e+06 W/m²
Surface heat balance error 0.0016 W/m²

Propellant Composition "Bas_4"

At pressure 1 MPa
Experimental burning rate 9.73 mm/s
Calculated burning rate 11.37 mm/s
First Conditional Fuel
Surface temperature 620.36 K
Linear burning rate 11.66 mm/s
Mass decomposition rate 19.35 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.48e+06 W/m²
Kinetic flame height 209.07 µm
Average kinetic flame density 1.83 kg/m³
Average kinetic flame temperature 2,163.68 K
Binder sublimation heat flux density 1.48e+06 W/m²
Surface heat balance error -4.37e-04 W/m²
Secondary Conditional Fuel
Surface temperature 600 K
Linear burning rate 3.99 mm/s
Mass decomposition rate 6.63 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 30,182.53 W/m²
Kinetic flame height 1,464.42 µm
Average kinetic flame density 4.83 kg/m³
Average kinetic flame temperature 821 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 104,134.85 W/m²
Kinetic flame height 615.55 µm
Average kinetic flame density 4.31 kg/m³
Average kinetic flame temperature 920.5 K
Heat flux density due to metal combustion, 74,413.46 W/m²
Effective thermal conductivity 1.41 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 1.66e-04 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 8.06e-05 m
Pore diameter 2.03e-06 m
Diffusion flame heat flux density 198,711.63 W/m²
Diffusion flame height 1,499.66 µm
Heat flux density to surface within skeletal structure 89,426.18 W/m²
Heat flux density to surface outside skeletal structure 15,065.57 W/m²
Total heat flux density to surface 303,203.37 W/m²
Binder sublimation heat flux density 303,203.37 W/m²
Surface heat balance error 1.63e-04 W/m²

At pressure 1.61 MPa
Experimental burning rate 13.58 mm/s
Calculated burning rate 13.89 mm/s
First Conditional Fuel

Surface temperature 630.61 K
Linear burning rate 19.49 mm/s
Mass decomposition rate 32.33 $kg{\cdot}s^{-1}{\cdot}m^{-2}$
Kinetic flame heat flux density to surface 2.96e+06 W/m^2
Kinetic flame height 103.8 μm
Average kinetic flame density 2.95 kg/m^3
Average kinetic flame temperature 2,168.81 K
Binder sublimation heat flux density 2.96e+06 W/m^2
Surface heat balance error -9.88e-04 W/m^2

Secondary Conditional Fuel

Surface temperature 602.39 K
Linear burning rate 4.55 mm/s
Mass decomposition rate 7.54 $kg{\cdot}s^{-1}{\cdot}m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 45,386.55 W/m^2
Kinetic flame height 968.59 μm
Average kinetic flame density 7.78 kg/m^3
Average kinetic flame temperature 822.19 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 261,309.68 W/m^2
Kinetic flame height 244.39 μm
Average kinetic flame density 6.94 kg/m^3
Average kinetic flame temperature 921.69 K
Heat flux density due to metal combustion, 88,205.94 W/m^2
Effective thermal conductivity 1.41 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 1.66e-04 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 6.79e-05 m
Pore diameter 2.01e-06 m

Diffusion flame heat flux density 174,438.48 W/m^2
Diffusion flame height 1,706.97 μm
Heat flux density to surface within skeletal structure 175,055.41 W/m^2
Heat flux density to surface outside skeletal structure 22,654.63 W/m^2
Total heat flux density to surface 372,148.52 W/m^2
Binder sublimation heat flux density 372,148.52 W/m^2
Surface heat balance error 7.25e-07 W/m^2

At pressure 2.22 MPa

Experimental burning rate 17.01 mm/s
Calculated burning rate 16.49 mm/s

First Conditional Fuel

Surface temperature 637.85 K
Linear burning rate 27.72 mm/s
Mass decomposition rate 45.98 $kg{\cdot}s^{-1}{\cdot}m^{-2}$
Kinetic flame heat flux density to surface 4.71e+06 W/m^2
Kinetic flame height 65.1 μm
Average kinetic flame density 4.06 kg/m^3
Average kinetic flame temperature 2,172.42 K
Binder sublimation heat flux density 4.71e+06 W/m^2
Surface heat balance error -0.0016 W/m^2

Secondary Conditional Fuel

Surface temperature 605.03 K
Linear burning rate 5.24 mm/s
Mass decomposition rate 8.69 $kg{\cdot}s^{-1}{\cdot}m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 56,663.64 W/m^2
Kinetic flame height 771.16 μm
Average kinetic flame density 10.71 kg/m^3

Average kinetic flame temperature 823.52 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 460,445.98 W/m^2

Kinetic flame height 138.12 μm

Average kinetic flame density 9.56 kg/m^3

Average kinetic flame temperature 923.02 K

Heat flux density due to metal combustion, 106,279.42 W/m^2

Effective thermal conductivity 1.41 W/(m·K)

Conductive thermal conductivity 1.41 W/(m·K)

Radiative thermal conductivity 1.66e-04 W/(m·K)

Conductive thermal conductivity balance error 9.240271880983641E-10

Skeleton layer thickness 5.62e-05 m

Pore diameter 1.99e-06 m

Diffusion flame heat flux density 151,219.89 W/m^2

Diffusion flame height 1,967.31 μm

Heat flux density to surface within skeletal structure 283,845.24 W/m^2

Heat flux density to surface outside skeletal structure 28,283.57 W/m^2

Total heat flux density to surface 463,348.7 W/m^2

Binder sublimation heat flux density 463,348.7 W/m^2

Surface heat balance error -1.99e-04 W/m^2

At pressure 2.83 MPa

Experimental burning rate 20.17 mm/s

Calculated burning rate 19.24 mm/s

First Conditional Fuel

Surface temperature 643.49 K

Linear burning rate 36.27 mm/s

Mass decomposition rate 60.17 kg·s^{-1}·m^{-2}

Kinetic flame heat flux density to surface 6.68e+06 W/m^2

Kinetic flame height 45.88 μm

Average kinetic flame density 5.17 kg/m^3

Average kinetic flame temperature 2,175.24 K

Binder sublimation heat flux density 6.68e+06 W/m^2

Surface heat balance error 5.65e-04 W/m^2

Secondary Conditional Fuel

Surface temperature 607.63 K

Linear burning rate 6.02 mm/s

Mass decomposition rate 9.98 kg·s^{-1}·m^{-2}

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 65,008.37 W/m^2

Kinetic flame height 668.18 μm

Average kinetic flame density 13.63 kg/m^3

Average kinetic flame temperature 824.81 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 685,055.66 W/m^2

Kinetic flame height 92.46 μm

Average kinetic flame density 12.17 kg/m^3

Average kinetic flame temperature 924.31 K

Heat flux density due to metal combustion, 127,446.21 W/m^2

Effective thermal conductivity 1.41 W/(m·K)

Conductive thermal conductivity 1.41 W/(m·K)

Radiative thermal conductivity 1.66e-04 W/(m·K)

Conductive thermal conductivity balance error 9.240271880983641E-10

Skeleton layer thickness 4.68e-05 m

Pore diameter 1.97e-06 m

Diffusion flame heat flux density 131,569.67 W/m^2

Diffusion flame height 2,259.16 μm

Heat flux density to surface within skeletal structure 406,942.74 W/m^2

Heat flux density to surface outside skeletal structure 32,448.83 W/m^2

Total heat flux density to surface 570,961.24 W/m²
Binder sublimation heat flux density 570,961.24 W/m²
Surface heat balance error -1.12e-04 W/m²

At pressure 3.44 MPa
Experimental burning rate 23.12 mm/s
Calculated burning rate 22.1 mm/s
First Conditional Fuel
Surface temperature 648.12 K
Linear burning rate 45.09 mm/s
Mass decomposition rate 74.81 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 8.82e+06 W/m²
Kinetic flame height 34.67 µm
Average kinetic flame density 6.28 kg/m³
Average kinetic flame temperature 2,177.56 K
Binder sublimation heat flux density 8.82e+06 W/m²
Surface heat balance error -0.0035 W/m²
Secondary Conditional Fuel
Surface temperature 610.06 K
Linear burning rate 6.84 mm/s
Mass decomposition rate 11.35 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 71,368.8 W/m²
Kinetic flame height 605.22 µm
Average kinetic flame density 16.55 kg/m³
Average kinetic flame temperature 826.03 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 926,659.55 W/m²
Kinetic flame height 68.09 µm
Average kinetic flame density 14.77 kg/m³
Average kinetic flame temperature 925.53 K
Heat flux density due to metal combustion, 150,916.48 W/m²
Effective thermal conductivity 1.41 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 1.67e-04 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 3.95e-05 m
Pore diameter 1.95e-06 m
Diffusion flame heat flux density 115,580.74 W/m²
Diffusion flame height 2,569.58 µm
Heat flux density to surface within skeletal structure 539,705.51 W/m²
Heat flux density to surface outside skeletal structure 35,623.63 W/m²
Total heat flux density to surface 690,909.89 W/m²
Binder sublimation heat flux density 690,909.89 W/m²
Surface heat balance error 2.22e-04 W/m²

At pressure 4.06 MPa
Experimental burning rate 25.92 mm/s
Calculated burning rate 25.03 mm/s
First Conditional Fuel
Surface temperature 652.07 K
Linear burning rate 54.15 mm/s
Mass decomposition rate 89.83 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.11e+07 W/m²
Kinetic flame height 27.46 µm
Average kinetic flame density 7.39 kg/m³
Average kinetic flame temperature 2,179.54 K
Binder sublimation heat flux density 1.11e+07 W/m²
Surface heat balance error 0.0033 W/m²

Secondary Conditional Fuel

Surface temperature 612.31 K
Linear burning rate 7.7 mm/s
Mass decomposition rate 12.78 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 76,393.34 W/m^2
Kinetic flame height 562.47 μm
Average kinetic flame density 19.46 kg/m^3
Average kinetic flame temperature 827.16 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1.18e+06 W/m^2
Kinetic flame height 53.24 μm
Average kinetic flame density 17.37 kg/m^3
Average kinetic flame temperature 926.66 K
Heat flux density due to metal combustion, 176,188.02 W/m^2
Effective thermal conductivity 1.41 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 1.67e-04 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 3.38e-05 m
Pore diameter 1.93e-06 m
Diffusion flame heat flux density 102,645.02 W/m^2
Diffusion flame height 2,891.21 μm
Heat flux density to surface within skeletal structure 679,715.44 W/m^2
Heat flux density to surface outside skeletal structure 38,131.62 W/m^2
Total heat flux density to surface 820,492.08 W/m^2
Binder sublimation heat flux density 820,492.09 W/m^2
Surface heat balance error -3.86e-04 W/m^2

At pressure 4.67 MPa

Experimental burning rate 28.6 mm/s
Calculated burning rate 28.02 mm/s

First Conditional Fuel

Surface temperature 655.52 K
Linear burning rate 63.41 mm/s
Mass decomposition rate 105.2 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.36e+07 W/m^2
Kinetic flame height 22.48 μm
Average kinetic flame density 8.49 kg/m^3
Average kinetic flame temperature 2,181.26 K
Binder sublimation heat flux density 1.36e+07 W/m^2
Surface heat balance error 0.003 W/m^2

Secondary Conditional Fuel

Surface temperature 614.38 K
Linear burning rate 8.58 mm/s
Mass decomposition rate 14.23 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 80,494.71 W/m^2
Kinetic flame height 531.24 μm
Average kinetic flame density 22.36 kg/m^3
Average kinetic flame temperature 828.19 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1.45e+06 W/m^2
Kinetic flame height 43.35 μm
Average kinetic flame density 19.97 kg/m^3
Average kinetic flame temperature 927.69 K
Heat flux density due to metal combustion, 202,934.63 W/m^2
Effective thermal conductivity 1.41 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)

Radiative thermal conductivity 1.67e-04 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 2.93e-05 m
Pore diameter 1.92e-06 m
Diffusion flame heat flux density 92,103.67 W/m²
Diffusion flame height 3,219.87 μm
Heat flux density to surface within skeletal structure 825,608.25 W/m²
Heat flux density to surface outside skeletal structure 40,178.82 W/m²
Total heat flux density to surface 957,890.73 W/m²
Binder sublimation heat flux density 957,890.74 W/m²
Surface heat balance error -1.97e-04 W/m²

At pressure 5.28 MPa
Experimental burning rate 31.17 mm/s
Calculated burning rate 31.03 mm/s
First Conditional Fuel
Surface temperature 658.58 K
Linear burning rate 72.87 mm/s
Mass decomposition rate 120.89 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.62e+07 W/m²
Kinetic flame height 18.87 μm
Average kinetic flame density 9.6 kg/m³
Average kinetic flame temperature 2,182.79 K
Binder sublimation heat flux density 1.62e+07 W/m²
Surface heat balance error -0.0041 W/m²
Secondary Conditional Fuel
Surface temperature 616.28 K
Linear burning rate 9.46 mm/s
Mass decomposition rate 15.7 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 83,936.01 W/m²
Kinetic flame height 507.19 μm
Average kinetic flame density 25.26 kg/m³
Average kinetic flame temperature 829.14 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1.72e+06 W/m²
Kinetic flame height 36.35 μm
Average kinetic flame density 22.56 kg/m³
Average kinetic flame temperature 928.64 K
Heat flux density due to metal combustion, 230,935.35 W/m²
Effective thermal conductivity 1.41 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 1.67e-04 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 2.57e-05 m
Pore diameter 1.9e-06 m
Diffusion flame heat flux density 83,412.85 W/m²
Diffusion flame height 3,553.07 μm
Heat flux density to surface within skeletal structure 976,532.02 W/m²
Heat flux density to surface outside skeletal structure 41,896.54 W/m²
Total heat flux density to surface 1.1e+06 W/m²
Binder sublimation heat flux density 1.1e+06 W/m²
Surface heat balance error -4.44e-04 W/m²

At pressure 5.89 MPa
Experimental burning rate 33.65 mm/s
Calculated burning rate 34.08 mm/s
First Conditional Fuel
Surface temperature 661.34 K

Linear burning rate 82.49 mm/s
Mass decomposition rate 136.85 kg·s^{-1}·m^{-2}
Kinetic flame heat flux density to surface 1.89e+07 W/m^2
Kinetic flame height 16.15 μm
Average kinetic flame density 10.7 kg/m^3
Average kinetic flame temperature 2,184.17 K
Binder sublimation heat flux density 1.89e+07 W/m^2
Surface heat balance error 2.82e-04 W/m^2

Secondary Conditional Fuel

Surface temperature 618.04 K
Linear burning rate 10.36 mm/s
Mass decomposition rate 17.19 kg·s^{-1}·m^{-2}

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 86,889.17 W/m^2
Kinetic flame height 487.93 μm
Average kinetic flame density 28.16 kg/m^3
Average kinetic flame temperature 830.02 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 2e+06 W/m^2
Kinetic flame height 31.15 μm
Average kinetic flame density 25.15 kg/m^3
Average kinetic flame temperature 929.52 K
Heat flux density due to metal combustion, 260,033.67 W/m^2
Effective thermal conductivity 1.41 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 1.67e-04 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 2.28e-05 m
Pore diameter 1.89e-06 m

Diffusion flame heat flux density 76,156.75 W/m^2
Diffusion flame height 3,889.29 μm
Heat flux density to surface within skeletal structure 1.13e+06 W/m^2
Heat flux density to surface outside skeletal structure 43,370.6 W/m^2
Total heat flux density to surface 1.25e+06 W/m^2
Binder sublimation heat flux density 1.25e+06 W/m^2
Surface heat balance error -4.98e-04 W/m^2

<u>At pressure 6.5 MPa</u>

Experimental burning rate 36.06 mm/s
Calculated burning rate 37.13 mm/s

First Conditional Fuel

Surface temperature 663.85 K
Linear burning rate 92.27 mm/s
Mass decomposition rate 153.08 kg·s^{-1}·m^{-2}
Kinetic flame heat flux density to surface 2.17e+07 W/m^2
Kinetic flame height 14.05 μm
Average kinetic flame density 11.8 kg/m^3
Average kinetic flame temperature 2,185.43 K
Binder sublimation heat flux density 2.17e+07 W/m^2
Surface heat balance error 0.0092 W/m^2

Secondary Conditional Fuel

Surface temperature 619.67 K
Linear burning rate 11.26 mm/s
Mass decomposition rate 18.68 kg·s^{-1}·m^{-2}

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 89,470.19 W/m^2
Kinetic flame height 472.03 μm
Average kinetic flame density 31.05 kg/m^3
Average kinetic flame temperature 830.84 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 2.29e+06 W/m²

Kinetic flame height 27.16 μm

Average kinetic flame density 27.73 kg/m³

Average kinetic flame temperature 930.34 K

Heat flux density due to metal combustion, 290,114 W/m²

Effective thermal conductivity 1.41 W/(m·K)

Conductive thermal conductivity 1.41 W/(m·K)

Radiative thermal conductivity 1.67e-04 W/(m·K)

Conductive thermal conductivity balance error 9.240271880983641E-10

Skeleton layer thickness 2.04e-05 m

Pore diameter 1.88e-06 m

Diffusion flame heat flux density 70,024.16 W/m²

Diffusion flame height 4,227.58 μm

Heat flux density to surface within skeletal structure 1.29e+06 W/m²

Heat flux density to surface outside skeletal structure 44,658.91 W/m²

Total heat flux density to surface 1.41e+06 W/m²

Binder sublimation heat flux density 1.41e+06 W/m²

Surface heat balance error -4.81e-04 W/m²

Propellant Composition "Bas_1"

At pressure 1 MPa

Experimental burning rate 14.26 mm/s

Calculated burning rate 15.19 mm/s

First Conditional Fuel

Surface temperature 627.58 K

Linear burning rate 10.98 mm/s

Mass decomposition rate 18.22 $kg·s^{-1}·m^{-2}$

Kinetic flame heat flux density to surface 1.59e+06 W/m²

Kinetic flame height 193.97 μm

Average kinetic flame density 1.83 kg/m³

Average kinetic flame temperature 2,167.29 K

Binder sublimation heat flux density 1.59e+06 W/m²

Surface heat balance error 8.02e-06 W/m²

Secondary Conditional Fuel

Surface temperature 621.8 K

Linear burning rate 8.18 mm/s

Mass decomposition rate 13.58 $kg·s^{-1}·m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 804,625.57 W/m²

Kinetic flame height 216.65 μm

Average kinetic flame density 2.66 kg/m³

Average kinetic flame temperature 1,493.4 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 229,345.85 W/m²

Kinetic flame height 422.59 μm

Average kinetic flame density 3.59 kg/m³

Average kinetic flame temperature 1,106.4 K

Heat flux density due to metal combustion, 773,826.2 W/m²

Effective thermal conductivity 0.4 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 2.17e-05 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 2.19e-06 m

Pore diameter 1.77e-07 m

Diffusion flame heat flux density 176,124.2 W/m²

Diffusion flame height 1,728.44 μm

Heat flux density to surface within skeletal structure 426,889.62 W/m²

Heat flux density to surface outside skeletal structure 462,225.37 W/m²
Total heat flux density to surface 1.07e+06 W/m²
Binder sublimation heat flux density 1.07e+06 W/m²
Surface heat balance error 3.05e-04 W/m²

At pressure 1.61 MPa
Experimental burning rate 19.92 mm/s
Calculated burning rate 20.2 mm/s
First Conditional Fuel
Surface temperature 638.11 K
Linear burning rate 18.51 mm/s
Mass decomposition rate 30.71 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 3.16e+06 W/m²
Kinetic flame height 97.09 μm
Average kinetic flame density 2.94 kg/m³
Average kinetic flame temperature 2,172.56 K
Binder sublimation heat flux density 3.16e+06 W/m²
Surface heat balance error 2.89e-04 W/m²
Secondary Conditional Fuel
Surface temperature 626.11 K
Linear burning rate 10.2 mm/s
Mass decomposition rate 16.92 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 1.1e+06 W/m²
Kinetic flame height 157.63 μm
Average kinetic flame density 4.28 kg/m³
Average kinetic flame temperature 1,495.56 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 550,431.38 W/m²
Kinetic flame height 175.3 μm
Average kinetic flame density 5.77 kg/m³
Average kinetic flame temperature 1,108.56 K
Heat flux density due to metal combustion, 1.03e+06 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 2.17e-05 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.64e-06 m
Pore diameter 1.74e-07 m
Diffusion flame heat flux density 141,137.07 W/m²
Diffusion flame height 2,153.85 μm
Heat flux density to surface within skeletal structure 635,273.64 W/m²
Heat flux density to surface outside skeletal structure 660,514.46 W/m²
Total heat flux density to surface 1.44e+06 W/m²
Binder sublimation heat flux density 1.44e+06 W/m²
Surface heat balance error 3.38e-04 W/m²

At pressure 2.22 MPa
Experimental burning rate 24.95 mm/s
Calculated burning rate 24.79 mm/s
First Conditional Fuel
Surface temperature 645.52 K
Linear burning rate 26.46 mm/s
Mass decomposition rate 43.89 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 5.01e+06 W/m²
Kinetic flame height 61.17 μm
Average kinetic flame density 4.05 kg/m³
Average kinetic flame temperature 2,176.26 K
Binder sublimation heat flux density 5.01e+06 W/m²

Surface heat balance error -9.88e-05 W/m²

Secondary Conditional Fuel

Surface temperature 629.52 K

Linear burning rate 12.11 mm/s

Mass decomposition rate 20.09 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.33e+06 W/m²

Kinetic flame height 130.16 µm

Average kinetic flame density 5.89 kg/m³

Average kinetic flame temperature 1,497.26 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 974,022.3 W/m²

Kinetic flame height 98.71 µm

Average kinetic flame density 7.94 kg/m³

Average kinetic flame temperature 1,110.26 K

Heat flux density due to metal combustion, 1.29e+06 W/m²

Effective thermal conductivity 0.4 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 2.18e-05 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 1.3e-06 m

Pore diameter 1.72e-07 m

Diffusion flame heat flux density 118,732.6 W/m²

Diffusion flame height 2,557.41 µm

Heat flux density to surface within skeletal structure 865,809.38 W/m²

Heat flux density to surface outside skeletal structure 824,348.64 W/m²

Total heat flux density to surface 1.81e+06 W/m²

Binder sublimation heat flux density 1.81e+06 W/m²

Surface heat balance error 2e-05 W/m²

At pressure 2.83 MPa

Experimental burning rate 29.57 mm/s

Calculated burning rate 29.13 mm/s

First Conditional Fuel

Surface temperature 651.28 K

Linear burning rate 34.73 mm/s

Mass decomposition rate 57.62 $kg \cdot s^{-1} \cdot m^{-2}$

Kinetic flame heat flux density to surface 7.07e+06 W/m²

Kinetic flame height 43.23 µm

Average kinetic flame density 5.16 kg/m³

Average kinetic flame temperature 2,179.14 K

Binder sublimation heat flux density 7.07e+06 W/m²

Surface heat balance error -0.0018 W/m²

Secondary Conditional Fuel

Surface temperature 632.35 K

Linear burning rate 13.94 mm/s

Mass decomposition rate 23.13 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.52e+06 W/m²

Kinetic flame height 113.85 µm

Average kinetic flame density 7.5 kg/m³

Average kinetic flame temperature 1,498.67 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.49e+06 W/m²

Kinetic flame height 64.54 µm

Average kinetic flame density 10.12 kg/m³

Average kinetic flame temperature 1,111.67 K

Heat flux density due to metal combustion, 1.56e+06 W/m²

Effective thermal conductivity 0.4 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 2.18e-05 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.08e-06 m
Pore diameter 1.7e-07 m
Diffusion flame heat flux density 103,043.16 W/m²
Diffusion flame height 2,944.06 µm
Heat flux density to surface within skeletal structure 1.11e+06 W/m²
Heat flux density to surface outside skeletal structure 965,794.13 W/m²
Total heat flux density to surface 2.18e+06 W/m²
Binder sublimation heat flux density 2.18e+06 W/m²
Surface heat balance error -7.88e-05 W/m²

At pressure 3.44 MPa
Experimental burning rate 33.9 mm/s
Calculated burning rate 33.29 mm/s
First Conditional Fuel
Surface temperature 656.02 K
Linear burning rate 43.28 mm/s
Mass decomposition rate 71.8 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 9.32e+06 W/m²
Kinetic flame height 32.74 µm
Average kinetic flame density 6.27 kg/m³
Average kinetic flame temperature 2,181.51 K
Binder sublimation heat flux density 9.32e+06 W/m²
Surface heat balance error 0.0038 W/m²
Secondary Conditional Fuel
Surface temperature 634.76 K
Linear burning rate 15.71 mm/s
Mass decomposition rate 26.06 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 1.68e+06 W/m²
Kinetic flame height 102.85 µm
Average kinetic flame density 9.11 kg/m³
Average kinetic flame temperature 1,499.88 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 2.07e+06 W/m²
Kinetic flame height 46.09 µm
Average kinetic flame density 12.28 kg/m³
Average kinetic flame temperature 1,112.88 K
Heat flux density due to metal combustion, 1.82e+06 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 2.18e-05 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 9.23e-07 m
Pore diameter 1.68e-07 m
Diffusion flame heat flux density 91,380.87 W/m²
Diffusion flame height 3,317.15 µm
Heat flux density to surface within skeletal structure 1.37e+06 W/m²
Heat flux density to surface outside skeletal structure 1.09e+06 W/m²
Total heat flux density to surface 2.55e+06 W/m²
Binder sublimation heat flux density 2.55e+06 W/m²
Surface heat balance error -9.24e-04 W/m²

At pressure 4.06 MPa
Experimental burning rate 38.01 mm/s
Calculated burning rate 37.31 mm/s
First Conditional Fuel

Surface temperature 660.05 K
Linear burning rate 52.06 mm/s
Mass decomposition rate 86.37 kg·s^{-1}·m^{-2}
Kinetic flame heat flux density to surface 1.17e+07 W/m^2
Kinetic flame height 25.97 μm
Average kinetic flame density 7.37 kg/m^3
Average kinetic flame temperature 2,183.52 K
Binder sublimation heat flux density 1.17e+07 W/m^2
Surface heat balance error 1.13e-05 W/m^2

Secondary Conditional Fuel

Surface temperature 636.87 K
Linear burning rate 17.42 mm/s
Mass decomposition rate 28.9 kg·s^{-1}·m^{-2}

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.82e+06 W/m^2
Kinetic flame height 94.82 μm
Average kinetic flame density 10.72 kg/m^3
Average kinetic flame temperature 1,500.93 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 2.73e+06 W/m^2
Kinetic flame height 34.89 μm
Average kinetic flame density 14.45 kg/m^3
Average kinetic flame temperature 1,113.93 K
Heat flux density due to metal combustion, 2.09e+06 W/m^2
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 2.18e-05 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 8.05e-07 m
Pore diameter 1.67e-07 m

Diffusion flame heat flux density 82,336.39 W/m^2
Diffusion flame height 3,678.97 μm
Heat flux density to surface within skeletal structure 1.63e+06 W/m^2
Heat flux density to surface outside skeletal structure 1.21e+06 W/m^2
Total heat flux density to surface 2.92e+06 W/m^2
Binder sublimation heat flux density 2.92e+06 W/m^2
Surface heat balance error -8.49e-04 W/m^2

<u>*At pressure 4.67 MPa*</u>

Experimental burning rate 41.93 mm/s
Calculated burning rate 41.22 mm/s

First Conditional Fuel

Surface temperature 663.56 K
Linear burning rate 61.06 mm/s
Mass decomposition rate 101.3 kg·s^{-1}·m^{-2}
Kinetic flame heat flux density to surface 1.43e+07 W/m^2
Kinetic flame height 21.29 μm
Average kinetic flame density 8.48 kg/m^3
Average kinetic flame temperature 2,185.28 K
Binder sublimation heat flux density 1.43e+07 W/m^2
Surface heat balance error -0.0032 W/m^2

Secondary Conditional Fuel

Surface temperature 638.74 K
Linear burning rate 19.09 mm/s
Mass decomposition rate 31.67 kg·s^{-1}·m^{-2}

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.95e+06 W/m^2
Kinetic flame height 88.63 μm
Average kinetic flame density 12.33 kg/m^3

Average kinetic flame temperature 1,501.87 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 3.46e+06 W/m^2

Kinetic flame height 27.51 μm

Average kinetic flame density 16.61 kg/m^3

Average kinetic flame temperature 1,114.87 K

Heat flux density due to metal combustion, 2.35e+06 W/m^2

Effective thermal conductivity 0.4 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 2.18e-05 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 7.13e-07 m

Pore diameter 1.66e-07 m

Diffusion flame heat flux density 75,096.12 W/m^2

Diffusion flame height 4,031.18 μm

Heat flux density to surface within skeletal structure 1.9e+06 W/m^2

Heat flux density to surface outside skeletal structure 1.31e+06 W/m^2

Total heat flux density to surface 3.29e+06 W/m^2

Binder sublimation heat flux density 3.29e+06 W/m^2

Surface heat balance error -7.79e-04 W/m^2

<u>*At pressure 5.28 MPa*</u>

Experimental burning rate 45.71 mm/s

Calculated burning rate 45.04 mm/s

First Conditional Fuel

Surface temperature 666.69 K

Linear burning rate 70.24 mm/s

Mass decomposition rate 116.54 kg·s^{-1}·m^{-2}

Kinetic flame heat flux density to surface 1.7e+07 W/m^2

Kinetic flame height 17.9 μm

Average kinetic flame density 9.58 kg/m^3

Average kinetic flame temperature 2,186.84 K

Binder sublimation heat flux density 1.7e+07 W/m^2

Surface heat balance error 0.0039 W/m^2

Secondary Conditional Fuel

Surface temperature 640.43 K

Linear burning rate 20.72 mm/s

Mass decomposition rate 34.37 kg·s^{-1}·m^{-2}

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 2.06e+06 W/m^2

Kinetic flame height 83.68 μm

Average kinetic flame density 13.94 kg/m^3

Average kinetic flame temperature 1,502.71 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 4.25e+06 W/m^2

Kinetic flame height 22.37 μm

Average kinetic flame density 18.77 kg/m^3

Average kinetic flame temperature 1,115.71 K

Heat flux density due to metal combustion, 2.62e+06 W/m^2

Effective thermal conductivity 0.4 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 2.18e-05 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 6.4e-07 m

Pore diameter 1.65e-07 m

Diffusion flame heat flux density 69,155.7 W/m^2

Diffusion flame height 4,375.02 μm

Heat flux density to surface within skeletal structure 2.18e+06 W/m^2

Heat flux density to surface outside skeletal structure 1.41e+06 W/m^2

Total heat flux density to surface 3.66e+06 W/m²
Binder sublimation heat flux density 3.66e+06 W/m²
Surface heat balance error 4.22e-04 W/m²

<u>At pressure 5.89 MPa</u>
Experimental burning rate 49.35 mm/s
Calculated burning rate 48.77 mm/s
First Conditional Fuel
Surface temperature 669.5 K
Linear burning rate 79.6 mm/s
Mass decomposition rate 132.06 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.98e+07 W/m²
Kinetic flame height 15.33 μm
Average kinetic flame density 10.68 kg/m³
Average kinetic flame temperature 2,188.25 K
Binder sublimation heat flux density 1.98e+07 W/m²
Surface heat balance error -0.0066 W/m²
Secondary Conditional Fuel
Surface temperature 641.96 K
Linear burning rate 22.31 mm/s
Mass decomposition rate 37.01 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 2.16e+06 W/m²
Kinetic flame height 79.59 μm
Average kinetic flame density 15.55 kg/m³
Average kinetic flame temperature 1,503.48 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 5.09e+06 W/m²
Kinetic flame height 18.63 μm
Average kinetic flame density 20.93 kg/m³
Average kinetic flame temperature 1,116.48 K
Heat flux density due to metal combustion, 2.88e+06 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 2.19e-05 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 5.8e-07 m
Pore diameter 1.64e-07 m
Diffusion flame heat flux density 64,184.93 W/m²
Diffusion flame height 4,711.45 μm
Heat flux density to surface within skeletal structure 2.46e+06 W/m²
Heat flux density to surface outside skeletal structure 1.5e+06 W/m²
Total heat flux density to surface 4.02e+06 W/m²
Binder sublimation heat flux density 4.02e+06 W/m²
Surface heat balance error 0.001 W/m²

<u>At pressure 6.5 MPa</u>
Experimental burning rate 52.88 mm/s
Calculated burning rate 52.44 mm/s
First Conditional Fuel
Surface temperature 672.06 K
Linear burning rate 89.12 mm/s
Mass decomposition rate 147.85 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 2.27e+07 W/m²
Kinetic flame height 13.34 μm
Average kinetic flame density 11.78 kg/m³
Average kinetic flame temperature 2,189.53 K
Binder sublimation heat flux density 2.27e+07 W/m²
Surface heat balance error 0.0063 W/m²

Secondary Conditional Fuel

Surface temperature 643.37 K
Linear burning rate 23.87 mm/s
Mass decomposition rate 39.6 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 2.26e+06 W/m²
Kinetic flame height 76.15 μm
Average kinetic flame density 17.15 kg/m³
Average kinetic flame temperature 1,504.18 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 6e+06 W/m²
Kinetic flame height 15.8 μm
Average kinetic flame density 23.09 kg/m³
Average kinetic flame temperature 1,117.18 K
Heat flux density due to metal combustion, 3.15e+06 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 2.19e-05 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 5.3e-07 m
Pore diameter 1.63e-07 m

Diffusion flame heat flux density 59,958.04 W/m²
Diffusion flame height 5,041.25 μm
Heat flux density to surface within skeletal structure 2.75e+06 W/m²
Heat flux density to surface outside skeletal structure 1.58e+06 W/m²
Total heat flux density to surface 4.39e+06 W/m²
Binder sublimation heat flux density 4.39e+06 W/m²
Surface heat balance error -3.51e-04 W/m²

Propellant Composition "Bas_0"

At pressure 1 MPa

Experimental burning rate 5.97 mm/s
Calculated burning rate 6 mm/s

First Conditional Fuel

Surface temperature 836.33 K
Linear burning rate 6.07 mm/s
Mass decomposition rate 10.07 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 4.03e+06 W/m²
Kinetic flame height 71.23 μm
Average kinetic flame density 1.75 kg/m³
Average kinetic flame temperature 2,271.67 K
Binder sublimation heat flux density 4.03e+06 W/m²
Surface heat balance error 0.0013 W/m²

Secondary Conditional Fuel

Surface temperature 818.14 K
Linear burning rate 2.96 mm/s
Mass decomposition rate 4.92 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 2.35e+06 W/m²
Kinetic flame height 65.74 μm
Average kinetic flame density 2.49 kg/m³
Average kinetic flame temperature 1,591.57 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 40,952.61 W/m²
Kinetic flame height 1,887.2 μm
Average kinetic flame density 3.29 kg/m³
Average kinetic flame temperature 1,204.57 K
Heat flux density due to metal combustion, 24,320.18 W/m²

Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 4.51e-04 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 6.15e-05 m
Pore diameter 1.78e-06 m
Diffusion flame heat flux density 454,856.71 W/m²
Diffusion flame height 626.1 μm
Heat flux density to surface within skeletal structure 27,776.17 W/m²
Heat flux density to surface outside skeletal structure 1.35e+06 W/m²
Total heat flux density to surface 1.83e+06 W/m²
Binder sublimation heat flux density 1.83e+06 W/m²
Surface heat balance error 4.36e-04 W/m²

At pressure 1.61 MPa
Experimental burning rate 7.9 mm/s
Calculated burning rate 7.88 mm/s
First Conditional Fuel
Surface temperature 851.65 K
Linear burning rate 10.84 mm/s
Mass decomposition rate 17.98 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 7.61e+06 W/m²
Kinetic flame height 37.52 μm
Average kinetic flame density 2.81 kg/m³
Average kinetic flame temperature 2,279.33 K
Binder sublimation heat flux density 7.61e+06 W/m²
Surface heat balance error -0.002 W/m²
Secondary Conditional Fuel
Surface temperature 823.6 K
Linear burning rate 3.69 mm/s
Mass decomposition rate 6.12 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 3.23e+06 W/m²
Kinetic flame height 47.72 μm
Average kinetic flame density 4.01 kg/m³
Average kinetic flame temperature 1,594.3 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 48,515.4 W/m²
Kinetic flame height 1,581.78 μm
Average kinetic flame density 5.3 kg/m³
Average kinetic flame temperature 1,207.3 K
Heat flux density due to metal combustion, 32,343.45 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 4.52e-04 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 4.61e-05 m
Pore diameter 1.75e-06 m
Diffusion flame heat flux density 365,023.44 W/m²
Diffusion flame height 778.69 μm
Heat flux density to surface within skeletal structure 32,442.68 W/m²
Heat flux density to surface outside skeletal structure 1.93e+06 W/m²
Total heat flux density to surface 2.33e+06 W/m²
Binder sublimation heat flux density 2.33e+06 W/m²
Surface heat balance error -4.35e-05 W/m²

At pressure 2.22 MPa
Experimental burning rate 9.56 mm/s
Calculated burning rate 9.51 mm/s

First Conditional Fuel

Surface temperature 862.32 K
Linear burning rate 16.03 mm/s
Mass decomposition rate 26.59 $kg\cdot s^{-1}\cdot m^{-2}$
Kinetic flame heat flux density to surface 1.17e+07 W/m^2
Kinetic flame height 24.36 μm
Average kinetic flame density 3.86 kg/m^3
Average kinetic flame temperature 2,284.66 K
Binder sublimation heat flux density 1.17e+07 W/m^2
Surface heat balance error -0.0021 W/m^2

Secondary Conditional Fuel

Surface temperature 827.67 K
Linear burning rate 4.33 mm/s
Mass decomposition rate 7.19 $kg\cdot s^{-1}\cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 3.94e+06 W/m^2
Kinetic flame height 38.99 μm
Average kinetic flame density 5.53 kg/m^3
Average kinetic flame temperature 1,596.34 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 53,778.79 W/m^2
Kinetic flame height 1,419.38 μm
Average kinetic flame density 7.29 kg/m^3
Average kinetic flame temperature 1,209.34 K
Heat flux density due to metal combustion, 39,933.71 W/m^2
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 4.52e-04 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 3.72e-05 m
Pore diameter 1.72e-06 m

Diffusion flame heat flux density 310,202.73 W/m^2
Diffusion flame height 914.99 μm
Heat flux density to surface within skeletal structure 35,774.69 W/m^2
Heat flux density to surface outside skeletal structure 2.44e+06 W/m^2
Total heat flux density to surface 2.78e+06 W/m^2
Binder sublimation heat flux density 2.78e+06 W/m^2
Surface heat balance error 4.82e-04 W/m^2

At pressure 2.83 MPa

Experimental burning rate 11.03 mm/s
Calculated burning rate 10.98 mm/s

First Conditional Fuel

Surface temperature 870.56 K
Linear burning rate 21.55 mm/s
Mass decomposition rate 35.75 $kg\cdot s^{-1}\cdot m^{-2}$
Kinetic flame heat flux density to surface 1.61e+07 W/m^2
Kinetic flame height 17.57 μm
Average kinetic flame density 4.91 kg/m^3
Average kinetic flame temperature 2,288.78 K
Binder sublimation heat flux density 1.61e+07 W/m^2
Surface heat balance error -0.0038 W/m^2

Secondary Conditional Fuel

Surface temperature 830.93 K
Linear burning rate 4.92 mm/s
Mass decomposition rate 8.17 $kg\cdot s^{-1}\cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 4.56e+06 W/m^2
Kinetic flame height 33.67 μm

Average kinetic flame density 7.04 kg/m^3
Average kinetic flame temperature 1,597.97 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 57,861.72 W/m^2
Kinetic flame height 1,313.59 μm
Average kinetic flame density 9.29 kg/m^3
Average kinetic flame temperature 1,210.97 K
Heat flux density due to metal combustion, 47,189.67 W/m^2
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 4.53e-04 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 3.14e-05 m
Pore diameter 1.71e-06 m
Diffusion flame heat flux density 272,684.02 W/m^2
Diffusion flame height 1,039.69 μm
Heat flux density to surface within skeletal structure 38,384.86 W/m^2
Heat flux density to surface outside skeletal structure 2.89e+06 W/m^2
Total heat flux density to surface 3.2e+06 W/m^2
Binder sublimation heat flux density 3.2e+06 W/m^2
Surface heat balance error -9.36e-04 W/m^2

At pressure 3.44 MPa
Experimental burning rate 12.38 mm/s
Calculated burning rate 12.34 mm/s
First Conditional Fuel
Surface temperature 877.3 K
Linear burning rate 27.34 mm/s
Mass decomposition rate 45.35 kg·s^{-1}·m^{-2}
Kinetic flame heat flux density to surface 2.09e+07 W/m^2
Kinetic flame height 13.51 μm
Average kinetic flame density 5.96 kg/m^3
Average kinetic flame temperature 2,292.15 K
Binder sublimation heat flux density 2.09e+07 W/m^2
Surface heat balance error -0.006 W/m^2
Secondary Conditional Fuel
Surface temperature 833.65 K
Linear burning rate 5.47 mm/s
Mass decomposition rate 9.08 kg·s^{-1}·m^{-2}
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 5.1e+06 W/m^2
Kinetic flame height 30 μm
Average kinetic flame density 8.55 kg/m^3
Average kinetic flame temperature 1,599.33 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 61,226.4 W/m^2
Kinetic flame height 1,236.97 μm
Average kinetic flame density 11.28 kg/m^3
Average kinetic flame temperature 1,212.33 K
Heat flux density due to metal combustion, 54,176.69 W/m^2
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 4.53e-04 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 2.73e-05 m
Pore diameter 1.69e-06 m
Diffusion flame heat flux density 245,103.06 W/m^2
Diffusion flame height 1,155.57 μm
Heat flux density to surface within skeletal structure 40,536.53 W/m^2

Heat flux density to surface outside skeletal structure 3.31e+06 W/m²
Total heat flux density to surface 3.6e+06 W/m²
Binder sublimation heat flux density 3.6e+06 W/m²
Surface heat balance error 2.7e-04 W/m²

At pressure 4.06 MPa
Experimental burning rate 13.63 mm/s
Calculated burning rate 13.62 mm/s
First Conditional Fuel
Surface temperature 883.03 K
Linear burning rate 33.36 mm/s
Mass decomposition rate 55.35 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 2.6e+07 W/m²
Kinetic flame height 10.85 μm
Average kinetic flame density 7.01 kg/m³
Average kinetic flame temperature 2,295.01 K
Binder sublimation heat flux density 2.6e+07 W/m²
Surface heat balance error -0.0077 W/m²
Secondary Conditional Fuel
Surface temperature 835.98 K
Linear burning rate 5.99 mm/s
Mass decomposition rate 9.93 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 5.6e+06 W/m²
Kinetic flame height 27.29 μm
Average kinetic flame density 10.06 kg/m³
Average kinetic flame temperature 1,600.49 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 64,107.25 W/m²
Kinetic flame height 1,177.75 μm
Average kinetic flame density 13.26 kg/m³
Average kinetic flame temperature 1,213.49 K
Heat flux density due to metal combustion, 60,939.85 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 4.54e-04 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 2.43e-05 m
Pore diameter 1.68e-06 m
Diffusion flame heat flux density 223,813.61 W/m²
Diffusion flame height 1,264.45 μm
Heat flux density to surface within skeletal structure 42,368.22 W/m²
Heat flux density to surface outside skeletal structure 3.7e+06 W/m²
Total heat flux density to surface 3.97e+06 W/m²
Binder sublimation heat flux density 3.97e+06 W/m²
Surface heat balance error 1.08e-04 W/m²

At pressure 4.67 MPa
Experimental burning rate 14.8 mm/s
Calculated burning rate 14.82 mm/s
First Conditional Fuel
Surface temperature 888.01 K
Linear burning rate 39.59 mm/s
Mass decomposition rate 65.68 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 3.14e+07 W/m²
Kinetic flame height 8.98 μm
Average kinetic flame density 8.06 kg/m³
Average kinetic flame temperature 2,297.5 K
Binder sublimation heat flux density 3.14e+07 W/m²

Surface heat balance error 0.0082 W/m²

Secondary Conditional Fuel

Surface temperature 838.02 K

Linear burning rate 6.48 mm/s

Mass decomposition rate 10.74 $kg\cdot s^{-1}\cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 6.06e+06 W/m²

Kinetic flame height 25.18 μm

Average kinetic flame density 11.57 kg/m³

Average kinetic flame temperature 1,601.51 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 66,639.29 W/m²

Kinetic flame height 1,129.94 μm

Average kinetic flame density 15.25 kg/m³

Average kinetic flame temperature 1,214.51 K

Heat flux density due to metal combustion, 67,511.83 W/m²

Effective thermal conductivity 0.4 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 4.54e-04 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 2.19e-05 m

Pore diameter 1.67e-06 m

Diffusion flame heat flux density 206,788 W/m²

Diffusion flame height 1,367.58 μm

Heat flux density to surface within skeletal structure 43,962.47 W/m²

Heat flux density to surface outside skeletal structure 4.08e+06 W/m²

Total heat flux density to surface 4.33e+06 W/m²

Binder sublimation heat flux density 4.33e+06 W/m²

Surface heat balance error -5.54e-04 W/m²

At pressure 5.28 MPa

Experimental burning rate 15.92 mm/s

Calculated burning rate 15.97 mm/s

First Conditional Fuel

Surface temperature 892.42 K

Linear burning rate 46 mm/s

Mass decomposition rate 76.32 $kg\cdot s^{-1}\cdot m^{-2}$

Kinetic flame heat flux density to surface 3.7e+07 W/m²

Kinetic flame height 7.61 μm

Average kinetic flame density 9.11 kg/m³

Average kinetic flame temperature 2,299.71 K

Binder sublimation heat flux density 3.7e+07 W/m²

Surface heat balance error 0.0042 W/m²

Secondary Conditional Fuel

Surface temperature 839.83 K

Linear burning rate 6.94 mm/s

Mass decomposition rate 11.51 $kg\cdot s^{-1}\cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 6.49e+06 W/m²

Kinetic flame height 23.49 μm

Average kinetic flame density 13.07 kg/m³

Average kinetic flame temperature 1,602.41 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 68,907.39 W/m²

Kinetic flame height 1,090.12 μm

Average kinetic flame density 17.23 kg/m³

Average kinetic flame temperature 1,215.41 K

Heat flux density due to metal combustion, 73,917.25 W/m²

Effective thermal conductivity 0.4 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 4.54e-04 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 2e-05 m
Pore diameter 1.66e-06 m
Diffusion flame heat flux density 192,801.25 W/m²
Diffusion flame height 1,465.85 μm
Heat flux density to surface within skeletal structure 45,372.75 W/m²
Heat flux density to surface outside skeletal structure 4.43e+06 W/m²
Total heat flux density to surface 4.67e+06 W/m²
Binder sublimation heat flux density 4.67e+06 W/m²
Surface heat balance error 5.65e-04 W/m²

At pressure 5.89 MPa
Experimental burning rate 16.98 mm/s
Calculated burning rate 17.06 mm/s
First Conditional Fuel
Surface temperature 896.39 K
Linear burning rate 52.59 mm/s
Mass decomposition rate 87.24 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 4.28e+07 W/m²
Kinetic flame height 6.57 μm
Average kinetic flame density 10.15 kg/m³
Average kinetic flame temperature 2,301.69 K
Binder sublimation heat flux density 4.28e+07 W/m²
Surface heat balance error -0.0043 W/m²
Secondary Conditional Fuel
Surface temperature 841.46 K
Linear burning rate 7.39 mm/s
Mass decomposition rate 12.25 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 6.9e+06 W/m²
Kinetic flame height 22.08 μm
Average kinetic flame density 14.58 kg/m³
Average kinetic flame temperature 1,603.23 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 70,968.39 W/m²
Kinetic flame height 1,056.16 μm
Average kinetic flame density 19.22 kg/m³
Average kinetic flame temperature 1,216.23 K
Heat flux density due to metal combustion, 80,175.33 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 4.54e-04 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.84e-05 m
Pore diameter 1.66e-06 m
Diffusion flame heat flux density 181,065.88 W/m²
Diffusion flame height 1,559.95 μm
Heat flux density to surface within skeletal structure 46,635.86 W/m²
Heat flux density to surface outside skeletal structure 4.77e+06 W/m²
Total heat flux density to surface 5e+06 W/m²
Binder sublimation heat flux density 5e+06 W/m²
Surface heat balance error 0.0014 W/m²

At pressure 6.5 MPa
Experimental burning rate 18 mm/s
Calculated burning rate 18.12 mm/s
First Conditional Fuel

Surface temperature 900 K
Linear burning rate 59.32 mm/s
Mass decomposition rate 98.41 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 4.88e+07 W/m^2
Kinetic flame height 5.75 μm
Average kinetic flame density 11.2 kg/m^3
Average kinetic flame temperature 2,303.5 K
Binder sublimation heat flux density 4.88e+07 W/m^2
Surface heat balance error 0.013 W/m^2

Secondary Conditional Fuel

Surface temperature 842.94 K
Linear burning rate 7.81 mm/s
Mass decomposition rate 12.96 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 7.29e+06 W/m^2
Kinetic flame height 20.89 μm
Average kinetic flame density 16.08 kg/m^3
Average kinetic flame temperature 1,603.97 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 72,862.23 W/m^2
Kinetic flame height 1,026.67 μm
Average kinetic flame density 21.2 kg/m^3
Average kinetic flame temperature 1,216.97 K
Heat flux density due to metal combustion, 86,301.43 W/m^2
Effective thermal conductivity 0.4 $W/(m \cdot K)$
Conductive thermal conductivity 0.4 $W/(m \cdot K)$
Radiative thermal conductivity 4.55e-04 $W/(m \cdot K)$
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.71e-05 m
Pore diameter 1.65e-06 m

Diffusion flame heat flux density 171,050.56 W/m^2
Diffusion flame height 1,650.42 μm
Heat flux density to surface within skeletal structure 47,778.34 W/m^2
Heat flux density to surface outside skeletal structure 5.1e+06 W/m^2
Total heat flux density to surface 5.32e+06 W/m^2
Binder sublimation heat flux density 5.32e+06 W/m^2
Surface heat balance error 0.0018 W/m^2

1 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	16.94 mm/s		15.24 mm/s		9.73 mm/s		14.26 mm/s		5.97 mm/s	
Calculated Burning Rate	14.36 mm/s		15.57 mm/s		11.37 mm/s		15.19 mm/s		6 mm/s	
Difference	2.59 mm/s		-0.33 mm/s		-1.65 mm/s		-0.93 mm/s		-0.038 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	11.66 mm/s	7.47 mm/s	11.66 mm/s	11.66 mm/s	11.66 mm/s	3.99 mm/s	10.98 mm/s	8.18 mm/s	6.07 mm/s	2.96 mm/s
Surface Temperature	620.36 K	611.74 K	620.36 K	620.36 K	620.36 K	600 K	627.58 K	621.8 K	836.33 K	818.14 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.48e+06 W/m²		1.48e+06 W/m²		1.48e+06 W/m²		1.59e+06 W/m²		4.03e+06 W/m²	
Kinetic Flame Height (Inter-Pocket)	209.07 µm		209.07 µm		209.07 µm		193.97 µm		71.23 µm	
Kinetic Flame Heat Flux Density (Skeleton)	167,628.46 W/m²		541,370.67 W/m²		104,134.85 W/m²		229,345.85 W/m²		40,952.61 W/m²	
Kinetic Flame Height (Skeleton)	584.18 µm		369.92 µm		615.55 µm		422.59 µm		1,887.2 µm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	877,543.79 W/m²		564,295.59 W/m²		30,182.53 W/m²		804,625.57 W/m²		2.35e+06 W/m²	
Kinetic Flame Height (Out-Skeleton)	199.79 µm		309.17 µm		1,464.42 µm		216.65 µm		65.74 µm	
Diffusion Flame Heat Flux Density	189,579.33 W/m²		903,760.77 W/m²		198,711.63 W/m²		176,124.2 W/m²		454,856.71 W/m²	
Diffusion Flame Height	1,578.37 µm		337.88 µm		1,499.66 µm		1,728.44 µm		626.1 µm	
Metal Combustion Heat Flux Density	48,498.9 W/m²		47,851.91 W/m²		74,413.46 W/m²		773,826.2 W/m²		24,320.18 W/m²	

1.61 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	21.71 mm/s		20.68 mm/s		13.58 mm/s		19.92 mm/s		7.9 mm/s	
Calculated Burning Rate	18.78 mm/s		20.67 mm/s		13.89 mm/s		20.2 mm/s		7.88 mm/s	
Difference	2.94 mm/s		0.012 mm/s		-0.3 mm/s		-0.29 mm/s		0.024 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	19.49 mm/s	9.22 mm/s	19.49 mm/s	13.4 mm/s	19.49 mm/s	4.55 mm/s	18.51 mm/s	10.2 mm/s	10.84 mm/s	3.69 mm/s
Surface Temperature	630.61 K	615.78 K	630.61 K	623.11 K	630.61 K	602.39 K	638.11 K	626.11 K	851.65 K	823.6 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	2.96e+06 W/m^2		2.96e+06 W/m^2		2.96e+06 W/m^2		3.16e+06 W/m^2		7.61e+06 W/m^2	
Kinetic Flame Height (Inter-Pocket)	103.8 μm		103.8 μm		103.8 μm		97.09 μm		37.52 μm	
Kinetic Flame Heat Flux Density (Skeleton)	388,002.6 W/m^2		1.34e+06 W/m^2		261,309.68 W/m^2		550,431.38 W/m^2		48,515.4 W/m^2	
Kinetic Flame Height (Skeleton)	251.34 μm		148.95 μm		244.39 μm		175.3 μm		1,581.78 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	1.21e+06 W/m^2		838,248.12 W/m^2		45,386.55 W/m^2		1.1e+06 W/m^2		3.23e+06 W/m^2	
Kinetic Flame Height (Out-Skeleton)	143.98 μm		207.8 μm		968.59 μm		157.63 μm		47.72 μm	
Diffusion Flame Heat Flux Density	153,433.03 W/m^2		785,612.93 W/m^2		174,438.48 W/m^2		141,137.07 W/m^2		365,023.44 W/m^2	
Diffusion Flame Height	1,947.57 μm		388.35 μm		1,706.97 μm		2,153.85 μm		778.69 μm	
Metal Combustion Heat Flux Density	63,908.82 W/m^2		57,442.72 W/m^2		88,205.94 W/m^2		1.03e+06 W/m^2		32,343.45 W/m^2	

2.22 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	25.67 mm/s		25.41 mm/s		17.01 mm/s		24.95 mm/s		9.56 mm/s	
Calculated Burning Rate	22.73 mm/s		25.19 mm/s		16.49 mm/s		24.79 mm/s		9.51 mm/s	
Difference	2.93 mm/s		0.22 mm/s		0.52 mm/s		0.15 mm/s		0.048 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	27.72 mm/s	10.84 mm/s	27.72 mm/s	15.21 mm/s	27.72 mm/s	5.24 mm/s	26.46 mm/s	12.11 mm/s	16.03 mm/s	4.33 mm/s
Surface Temperature	637.85 K	618.93 K	637.85 K	625.62 K	637.85 K	605.03 K	645.52 K	629.52 K	862.32 K	827.67 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	4.71e+06 W/m^2		4.71e+06 W/m^2		4.71e+06 W/m^2		5.01e+06 W/m^2		1.17e+07 W/m^2	
Kinetic Flame Height (Inter-Pocket)	65.1 μm		65.1 μm		65.1 μm		61.17 μm		24.36 μm	
Kinetic Flame Heat Flux Density (Skeleton)	669,934.27 W/m^2		2.4e+06 W/m^2		460,445.98 W/m^2		974,022.3 W/m^2		53,778.79 W/m^2	
Kinetic Flame Height (Skeleton)	145.1 μm		83.28 μm		138.12 μm		98.71 μm		1,419.38 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	1.48e+06 W/m^2		1.06e+06 W/m^2		56,663.64 W/m^2		1.33e+06 W/m^2		3.94e+06 W/m^2	
Kinetic Flame Height (Out-Skeleton)	117.71 μm		164.09 μm		771.16 μm		130.16 μm		38.99 μm	
Diffusion Flame Heat Flux Density	130,397.51 W/m^2		691,875.21 W/m^2		151,219.89 W/m^2		118,732.6 W/m^2		310,202.73 W/m^2	
Diffusion Flame Height	2,289.21 μm		440.6 μm		1,967.31 μm		2,557.41 μm		914.99 μm	
Metal Combustion Heat Flux Density	79,011.49 W/m^2		67,791.41 W/m^2		106,279.42 W/m^2		1.29e+06 W/m^2		39,933.71 W/m^2	

2.83 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	29.12 mm/s		29.68 mm/s		20.17 mm/s		29.57 mm/s		11.03 mm/s	
Calculated Burning Rate	26.41 mm/s		29.42 mm/s		19.24 mm/s		29.13 mm/s		10.98 mm/s	
Difference	2.71 mm/s		0.26 mm/s		0.93 mm/s		0.44 mm/s		0.048 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	36.27 mm/s	12.36 mm/s	36.27 mm/s	17 mm/s	36.27 mm/s	6.02 mm/s	34.73 mm/s	13.94 mm/s	21.55 mm/s	4.92 mm/s
Surface Temperature	643.49 K	621.51 K	643.49 K	627.86 K	643.49 K	607.63 K	651.28 K	632.35 K	870.56 K	830.93 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	6.68e+06 W/m^2		6.68e+06 W/m^2		6.68e+06 W/m^2		7.07e+06 W/m^2		1.61e+07 W/m^2	
Kinetic Flame Height (Inter-Pocket)	45.88 μm		45.88 μm		45.88 μm		43.23 μm		17.57 μm	
Kinetic Flame Heat Flux Density (Skeleton)	1e+06 W/m^2		3.66e+06 W/m^2		685,055.66 W/m^2		1.49e+06 W/m^2		57,861.72 W/m^2	
Kinetic Flame Height (Skeleton)	96.66 μm		54.55 μm		92.46 μm		64.54 μm		1,313.59 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	1.71e+06 W/m^2		1.25e+06 W/m^2		65,008.37 W/m^2		1.52e+06 W/m^2		4.56e+06 W/m^2	
Kinetic Flame Height (Out-Skeleton)	102.01 μm		139.47 μm		668.18 μm		113.85 μm		33.67 μm	
Diffusion Flame Heat Flux Density	114,258.94 W/m^2		618,381.64 W/m^2		131,569.67 W/m^2		103,043.16 W/m^2		272,684.02 W/m^2	
Diffusion Flame Height	2,610.29 μm		492.6 μm		2,259.16 μm		2,944.06 μm		1,039.69 μm	
Metal Combustion Heat Flux Density	93,864.73 W/m^2		78,478.26 W/m^2		127,446.21 W/m^2		1.56e+06 W/m^2		47,189.67 W/m^2	

3.44 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	32.23 mm/s		33.63 mm/s		23.12 mm/s		33.9 mm/s		12.38 mm/s	
Calculated Burning Rate	29.88 mm/s		33.45 mm/s		22.1 mm/s		33.29 mm/s		12.34 mm/s	
Difference	2.35 mm/s		0.19 mm/s		1.02 mm/s		0.62 mm/s		0.035 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	45.09 mm/s	13.8 mm/s	45.09 mm/s	18.76 mm/s	45.09 mm/s	6.84 mm/s	43.28 mm/s	15.71 mm/s	27.34 mm/s	5.47 mm/s
Surface Temperature	648.12 K	623.69 K	648.12 K	629.85 K	648.12 K	610.06 K	656.02 K	634.76 K	877.3 K	833.65 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	8.82e+06 W/m²		8.82e+06 W/m²		8.82e+06 W/m²		9.32e+06 W/m²		2.09e+07 W/m²	
Kinetic Flame Height (Inter-Pocket)	34.67 μm		34.67 μm		34.67 μm		32.74 μm		13.51 μm	
Kinetic Flame Heat Flux Density (Skeleton)	1.38e+06 W/m²		5.09e+06 W/m²		926,659.55 W/m²		2.07e+06 W/m²		61,226.4 W/m²	
Kinetic Flame Height (Skeleton)	70.07 μm		39.16 μm		68.09 μm		46.09 μm		1,236.97 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	1.91e+06 W/m²		1.41e+06 W/m²		71,368.8 W/m²		1.68e+06 W/m²		5.1e+06 W/m²	
Kinetic Flame Height (Out-Skeleton)	91.36 μm		123.45 μm		605.22 μm		102.85 μm		30 μm	
Diffusion Flame Heat Flux Density	102,230.23 W/m²		560,009.36 W/m²		115,580.74 W/m²		91,380.87 W/m²		245,103.06 W/m²	
Diffusion Flame Height	2,915.29 μm		543.59 μm		2,569.58 μm		3,317.15 μm		1,155.57 μm	
Metal Combustion Heat Flux Density	108,513.21 W/m²		89,304.68 W/m²		150,916.48 W/m²		1.82e+06 W/m²		54,176.69 W/m²	

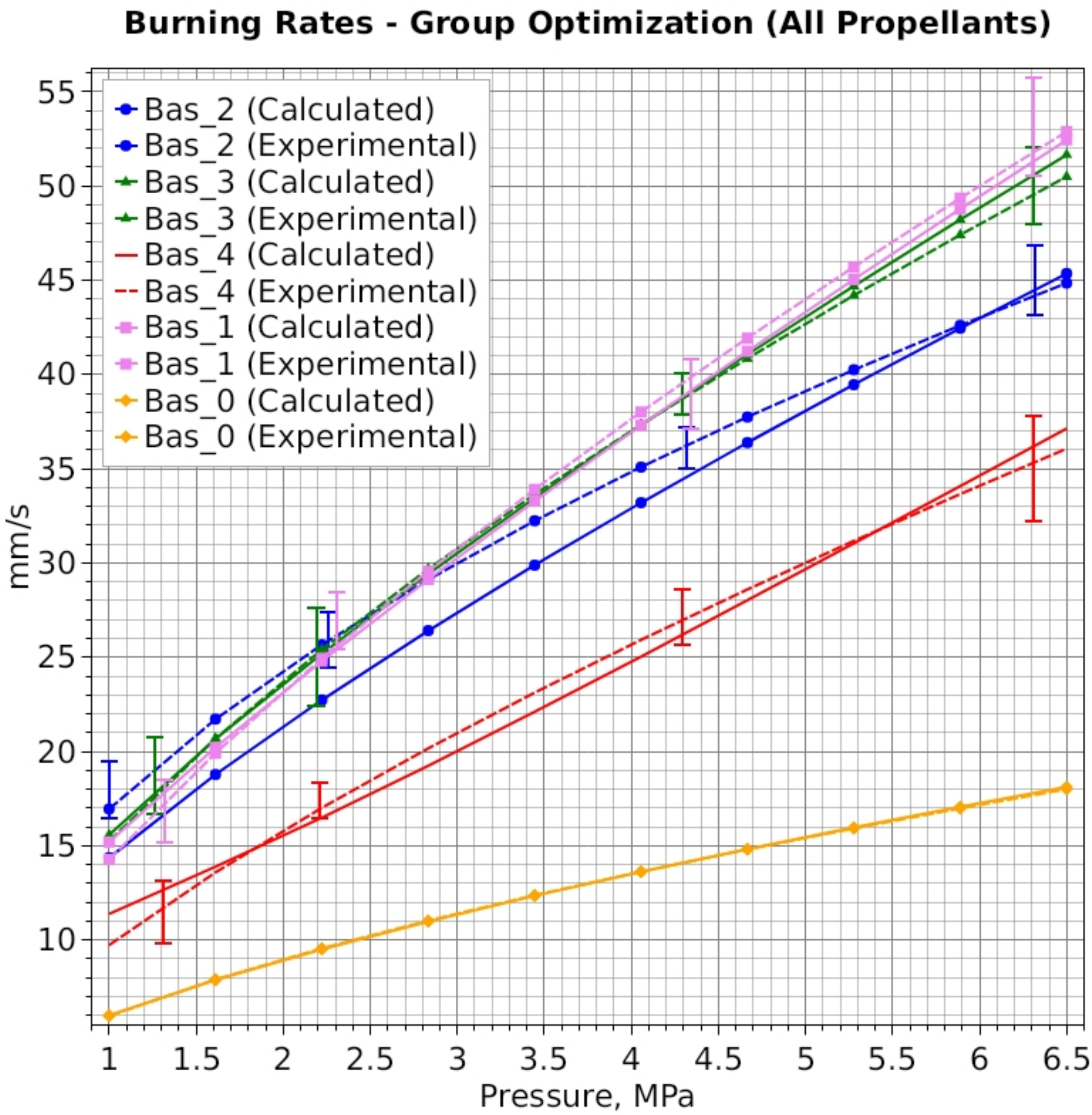
4.06 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	35.09 mm/s		37.34 mm/s		25.92 mm/s		38.01 mm/s		13.63 mm/s	
Calculated Burning Rate	33.2 mm/s		37.32 mm/s		25.03 mm/s		37.31 mm/s		13.62 mm/s	
Difference	1.89 mm/s		0.021 mm/s		0.89 mm/s		0.7 mm/s		0.013 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	54.15 mm/s	15.19 mm/s	54.15 mm/s	20.48 mm/s	54.15 mm/s	7.7 mm/s	52.06 mm/s	17.42 mm/s	33.36 mm/s	5.99 mm/s
Surface Temperature	652.07 K	625.59 K	652.07 K	631.63 K	652.07 K	612.31 K	660.05 K	636.87 K	883.03 K	835.98 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.11e+07 W/m^2		1.11e+07 W/m^2		1.11e+07 W/m^2		1.17e+07 W/m^2		2.6e+07 W/m^2	
Kinetic Flame Height (Inter-Pocket)	27.46 µm		27.46 µm		27.46 µm		25.97 µm		10.85 µm	
Kinetic Flame Heat Flux Density (Skeleton)	1.8e+06 W/m^2		6.67e+06 W/m^2		1.18e+06 W/m^2		2.73e+06 W/m^2		64,107.25 W/m^2	
Kinetic Flame Height (Skeleton)	53.7 µm		29.83 µm		53.24 µm		34.89 µm		1,177.75 µm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	2.08e+06 W/m^2		1.55e+06 W/m^2		76,393.34 W/m^2		1.82e+06 W/m^2		5.6e+06 W/m^2	
Kinetic Flame Height (Out-Skeleton)	83.56 µm		112.04 µm		562.47 µm		94.82 µm		27.29 µm	
Diffusion Flame Heat Flux Density	92,866.51 W/m^2		512,780.65 W/m^2		102,645.02 W/m^2		82,336.39 W/m^2		223,813.61 W/m^2	
Diffusion Flame Height	3,207.19 µm		593.31 µm		2,891.21 µm		3,678.97 µm		1,264.45 µm	
Metal Combustion Heat Flux Density	122,988.48 W/m^2		100,170.66 W/m^2		176,188.02 W/m^2		2.09e+06 W/m^2		60,939.85 W/m^2	

4.67 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	37.75 mm/s		40.85 mm/s		28.6 mm/s		41.93 mm/s		14.8 mm/s	
Calculated Burning Rate	36.39 mm/s		41.06 mm/s		28.02 mm/s		41.22 mm/s		14.82 mm/s	
Difference	1.36 mm/s		-0.21 mm/s		0.58 mm/s		0.71 mm/s		-0.015 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	63.41 mm/s	16.52 mm/s	63.41 mm/s	22.15 mm/s	63.41 mm/s	8.58 mm/s	61.06 mm/s	19.09 mm/s	39.59 mm/s	6.48 mm/s
Surface Temperature	655.52 K	627.28 K	655.52 K	633.23 K	655.52 K	614.38 K	663.56 K	638.74 K	888.01 K	838.02 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.36e+07 W/m²		1.36e+07 W/m²		1.36e+07 W/m²		1.43e+07 W/m²		3.14e+07 W/m²	
Kinetic Flame Height (Inter-Pocket)	22.48 µm		22.48 µm		22.48 µm		21.29 µm		8.98 µm	
Kinetic Flame Heat Flux Density (Skeleton)	2.25e+06 W/m²		8.4e+06 W/m²		1.45e+06 W/m²		3.46e+06 W/m²		66,639.29 W/m²	
Kinetic Flame Height (Skeleton)	42.8 µm		23.69 µm		43.35 µm		27.51 µm		1,129.94 µm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	2.24e+06 W/m²		1.67e+06 W/m²		80,494.71 W/m²		1.95e+06 W/m²		6.06e+06 W/m²	
Kinetic Flame Height (Out-Skeleton)	77.54 µm		103.4 µm		531.24 µm		88.63 µm		25.18 µm	
Diffusion Flame Heat Flux Density	85,338.99 W/m²		473,858.26 W/m²		92,103.67 W/m²		75,096.12 W/m²		206,788 W/m²	
Diffusion Flame Height	3,488.12 µm		641.71 µm		3,219.87 µm		4,031.18 µm		1,367.58 µm	
Metal Combustion Heat Flux Density	137,313.34 W/m²		111,023.52 W/m²		202,934.63 W/m²		2.35e+06 W/m²		67,511.83 W/m²	

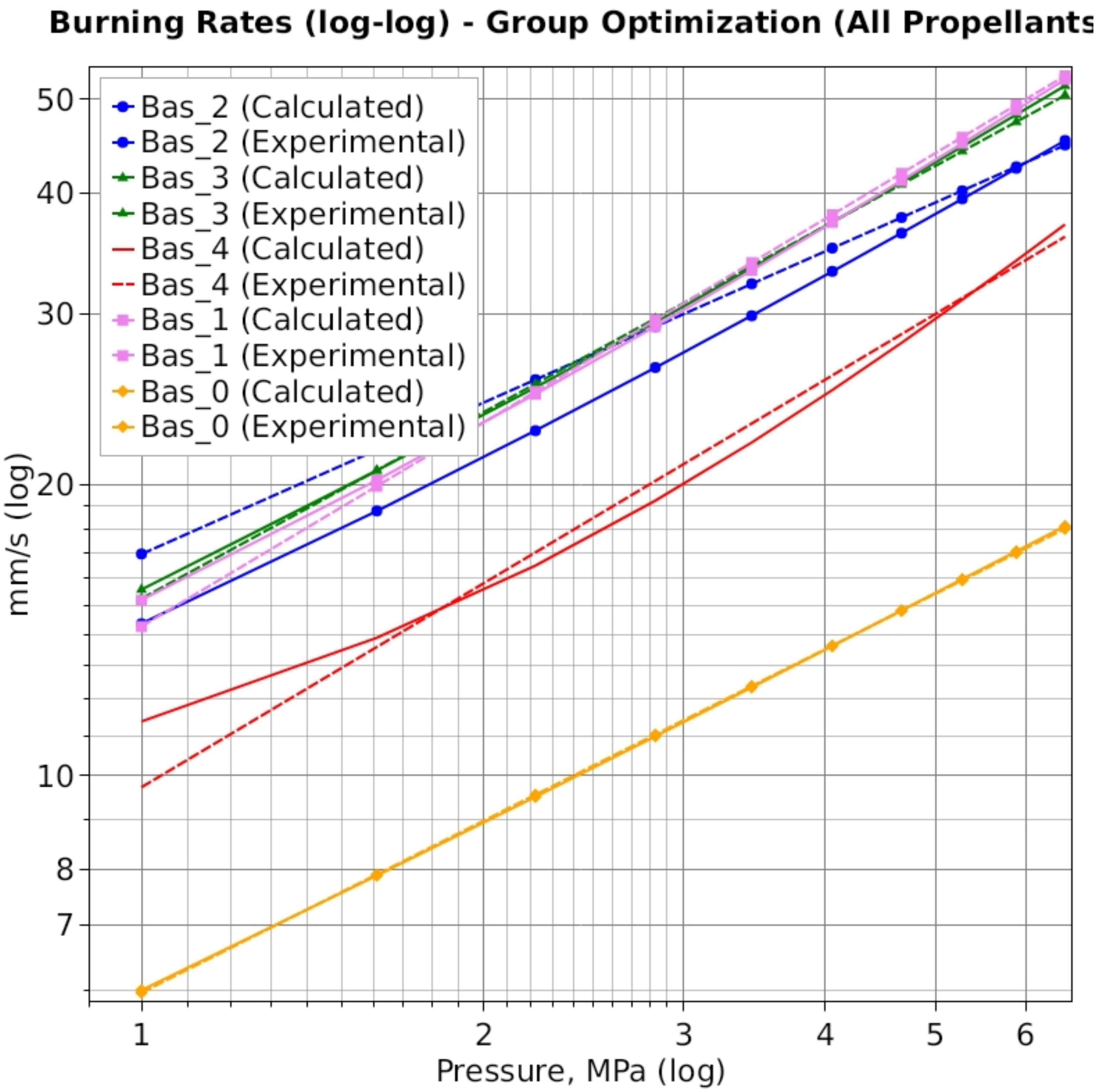
5.28 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	40.24 mm/s		44.2 mm/s		31.17 mm/s		45.71 mm/s		15.92 mm/s	
Calculated Burning Rate	39.46 mm/s		44.69 mm/s		31.03 mm/s		45.04 mm/s		15.97 mm/s	
Difference	0.78 mm/s		-0.49 mm/s		0.14 mm/s		0.67 mm/s		-0.046 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	72.87 mm/s	17.8 mm/s	72.87 mm/s	23.78 mm/s	72.87 mm/s	9.46 mm/s	70.24 mm/s	20.72 mm/s	46 mm/s	6.94 mm/s
Surface Temperature	658.58 K	628.78 K	658.58 K	634.68 K	658.58 K	616.28 K	666.69 K	640.43 K	892.42 K	839.83 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.62e+07 W/m²		1.62e+07 W/m²		1.62e+07 W/m²		1.7e+07 W/m²		3.7e+07 W/m²	
Kinetic Flame Height (Inter-Pocket)	18.87 µm		18.87 µm		18.87 µm		17.9 µm		7.61 µm	
Kinetic Flame Heat Flux Density (Skeleton)	2.74e+06 W/m²		1.03e+07 W/m²		1.72e+06 W/m²		4.25e+06 W/m²		68,907.39 W/m²	
Kinetic Flame Height (Skeleton)	35.13 µm		19.39 µm		36.35 µm		22.37 µm		1,090.12 µm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	2.39e+06 W/m²		1.79e+06 W/m²		83,936.01 W/m²		2.06e+06 W/m²		6.49e+06 W/m²	
Kinetic Flame Height (Out-Skeleton)	72.72 µm		96.57 µm		507.19 µm		83.68 µm		23.49 µm	
Diffusion Flame Heat Flux Density	79,135.68 W/m²		441,241.32 W/m²		83,412.85 W/m²		69,155.7 W/m²		192,801.25 W/m²	
Diffusion Flame Height	3,759.64 µm		688.81 µm		3,553.07 µm		4,375.02 µm		1,465.85 µm	
Metal Combustion Heat Flux Density	151,504.8 W/m²		121,834.82 W/m²		230,935.35 W/m²		2.62e+06 W/m²		73,917.25 W/m²	

5.89 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	42.6 mm/s		47.41 mm/s		33.65 mm/s		49.35 mm/s		16.98 mm/s	
Calculated Burning Rate	42.45 mm/s		48.21 mm/s		34.08 mm/s		48.77 mm/s		17.06 mm/s	
Difference	0.15 mm/s		-0.81 mm/s		-0.42 mm/s		0.58 mm/s		-0.08 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	82.49 mm/s	19.05 mm/s	82.49 mm/s	25.36 mm/s	82.49 mm/s	10.36 mm/s	79.6 mm/s	22.31 mm/s	52.59 mm/s	7.39 mm/s
Surface Temperature	661.34 K	630.15 K	661.34 K	636.01 K	661.34 K	618.04 K	669.5 K	641.96 K	896.39 K	841.46 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.89e+07 W/m²		1.89e+07 W/m²		1.89e+07 W/m²		1.98e+07 W/m²		4.28e+07 W/m²	
Kinetic Flame Height (Inter-Pocket)	16.15 μm		16.15 μm		16.15 μm		15.33 μm		6.57 μm	
Kinetic Flame Heat Flux Density (Skeleton)	3.26e+06 W/m²		1.22e+07 W/m²		2e+06 W/m²		5.09e+06 W/m²		70,968.39 W/m²	
Kinetic Flame Height (Skeleton)	29.49 μm		16.25 μm		31.15 μm		18.63 μm		1,056.16 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	2.52e+06 W/m²		1.9e+06 W/m²		86,889.17 W/m²		2.16e+06 W/m²		6.9e+06 W/m²	
Kinetic Flame Height (Out-Skeleton)	68.74 μm		91 μm		487.93 μm		79.59 μm		22.08 μm	
Diffusion Flame Heat Flux Density	73,921.95 W/m²		413,503.88 W/m²		76,156.75 W/m²		64,184.93 W/m²		181,065.88 W/m²	
Diffusion Flame Height	4,022.95 μm		734.69 μm		3,889.29 μm		4,711.45 μm		1,559.95 μm	
Metal Combustion Heat Flux Density	165,576.03 W/m²		132,589.21 W/m²		260,033.67 W/m²		2.88e+06 W/m²		80,175.33 W/m²	

6.5 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	44.85 mm/s		50.5 mm/s		36.06 mm/s		52.88 mm/s		18 mm/s	
Calculated Burning Rate	45.35 mm/s		51.65 mm/s		37.13 mm/s		52.44 mm/s		18.12 mm/s	
Difference	-0.5 mm/s		-1.15 mm/s		-1.07 mm/s		0.44 mm/s		-0.11 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	92.27 mm/s	20.26 mm/s	92.27 mm/s	26.9 mm/s	92.27 mm/s	11.26 mm/s	89.12 mm/s	23.87 mm/s	59.32 mm/s	7.81 mm/s
Surface Temperature	663.85 K	631.41 K	663.85 K	637.23 K	663.85 K	619.67 K	672.06 K	643.37 K	900 K	842.94 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	2.17e+07 W/m²		2.17e+07 W/m²		2.17e+07 W/m²		2.27e+07 W/m²		4.88e+07 W/m²	
Kinetic Flame Height (Inter-Pocket)	14.05 µm		14.05 µm		14.05 µm		13.34 µm		5.75 µm	
Kinetic Flame Heat Flux Density (Skeleton)	3.81e+06 W/m²		1.43e+07 W/m²		2.29e+06 W/m²		6e+06 W/m²		72,862.23 W/m²	
Kinetic Flame Height (Skeleton)	25.21 µm		13.87 µm		27.16 µm		15.8 µm		1,026.67 µm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	2.65e+06 W/m²		2e+06 W/m²		89,470.19 W/m²		2.26e+06 W/m²		7.29e+06 W/m²	
Kinetic Flame Height (Out-Skeleton)	65.38 µm		86.34 µm		472.03 µm		76.15 µm		20.89 µm	
Diffusion Flame Heat Flux Density	69,469.11 W/m²		389,611.96 W/m²		70,024.16 W/m²		59,958.04 W/m²		171,050.56 W/m²	
Diffusion Flame Height	4,279.01 µm		779.43 µm		4,227.58 µm		5,041.25 µm		1,650.42 µm	
Metal Combustion Heat Flux Density	179,537.53 W/m²		143,278.68 W/m²		290,114 W/m²		3.15e+06 W/m²		86,301.43 W/m²	



Burning Rates - Group Optimization (All Propellants)
Bas_2 (Calculated)
Bas_2 (Experimental)
Bas_3 (Calculated)
Bas_3 (Experimental)
Bas_4 (Calculated)
Bas_4 (Experimental)
Bas_1 (Calculated)
Bas_1 (Experimental)
Bas_0 (Calculated)
Bas_0 (Experimental)
mm/s
Pressure, MPa
55
50
45
40
35
30
25
20
15
10
1
1.5
2
2.5
3
3.5
4
4.5
5
5.5
6
6.5



Propellant Composition Parameters

Composition "Bas_2"

Density 1,659 kg/m^3, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: U = A · p^v, A = 1.285341E-05; v = 0.52.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Diffusion flame temperature 3,604 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)

Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K

Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_3"

Density 1,659 kg/m^3, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.57.

Vieille's Power Law: U = A · p^v, A = 2.203002E-06; v = 0.64.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,674 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,623 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_4"

Density 1,659 kg/m^3, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.56.

Vieille's Power Law: U = A · p^v, A = 6.137894E-07; v = 0.7.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,580 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,241 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,042 K

Composition "Bas_1"

Density 1,659 kg/m^3, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: U = A · p^v, A = 9.000448E-07; v = 0.7.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_0"

Density 1,659 kg/m^3, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.
Vieille's Power Law: U = A · p^v, A = 1.720582E-06; v = 0.59.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

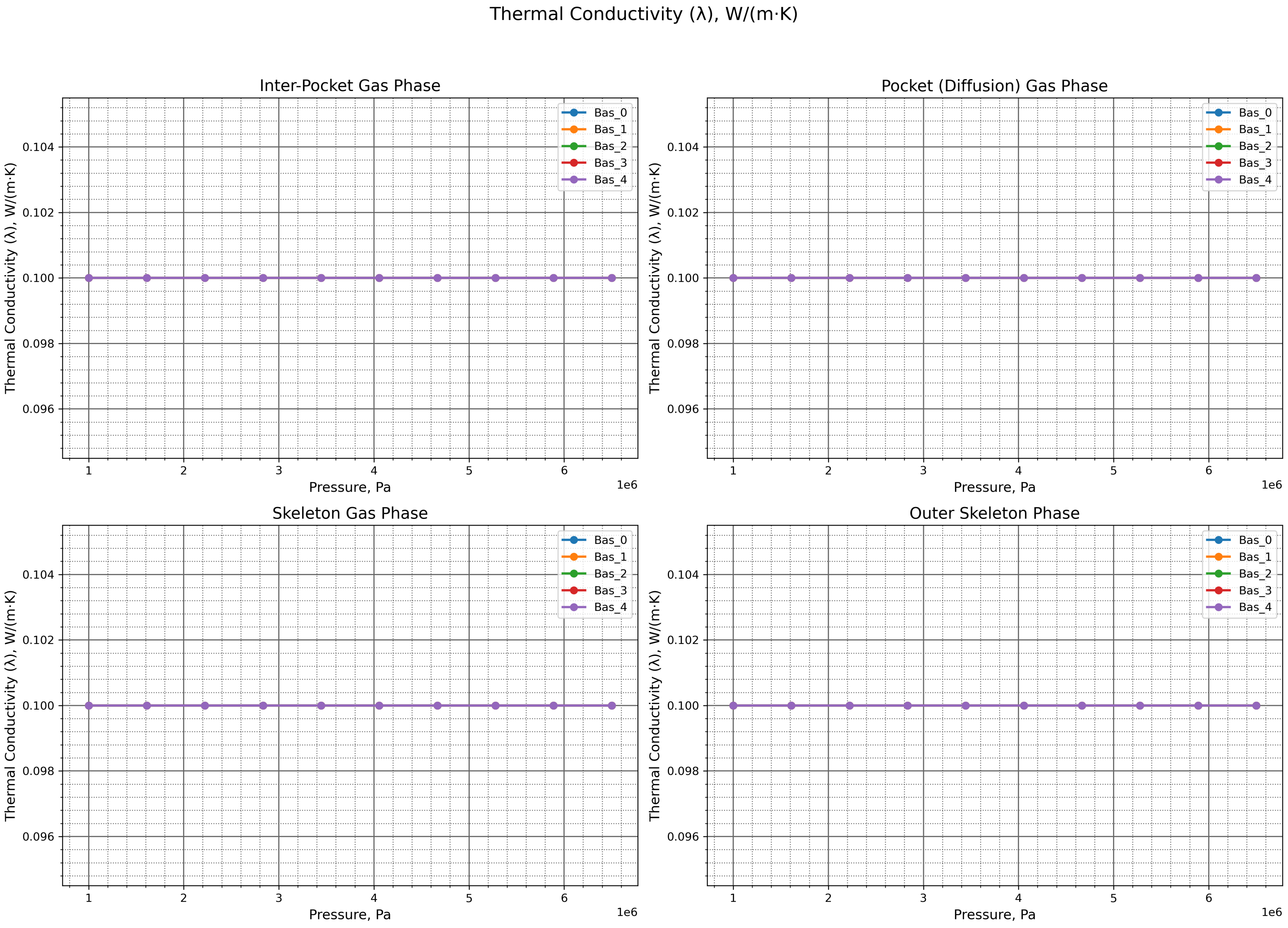
Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

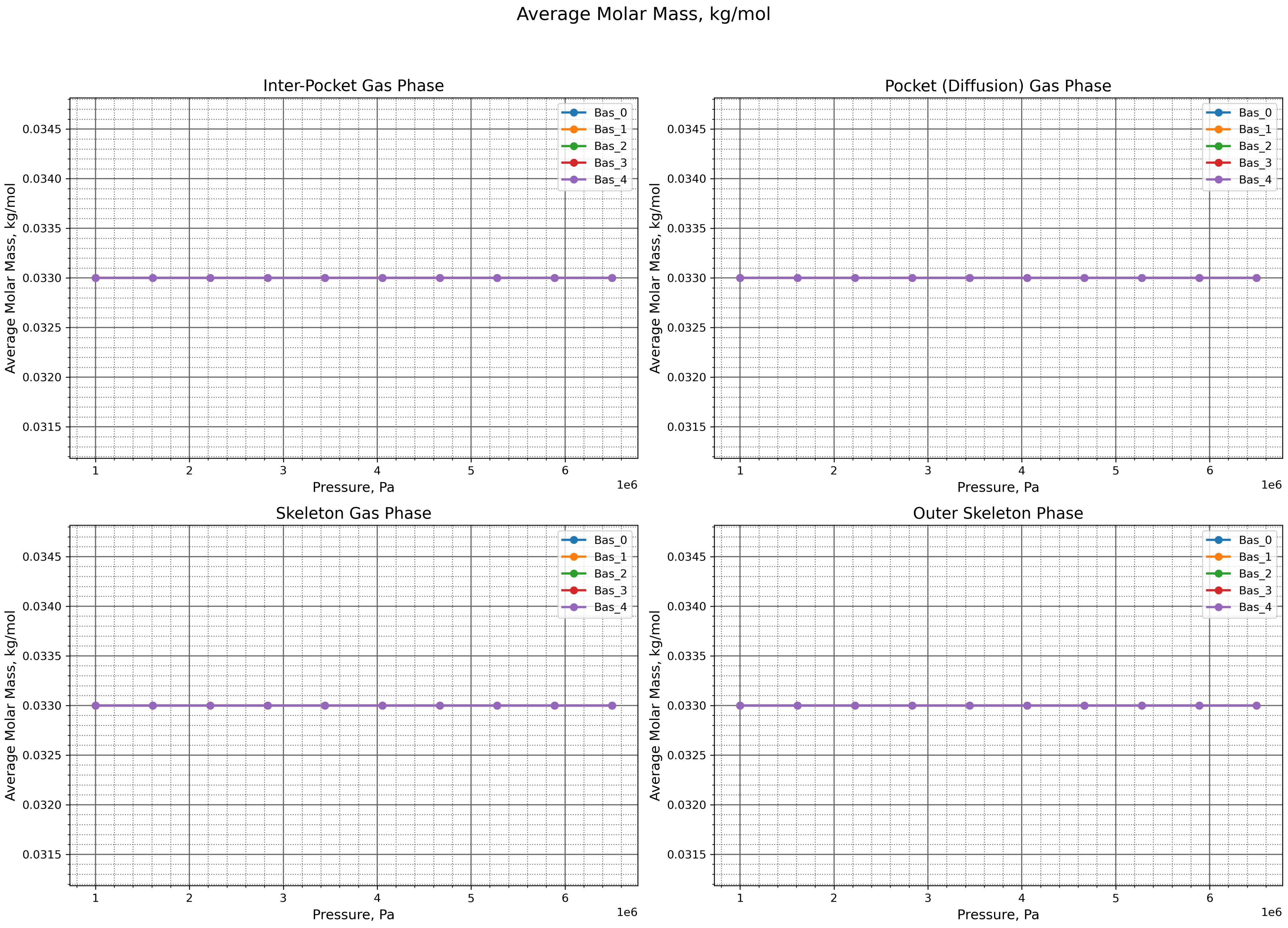
Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,591 K

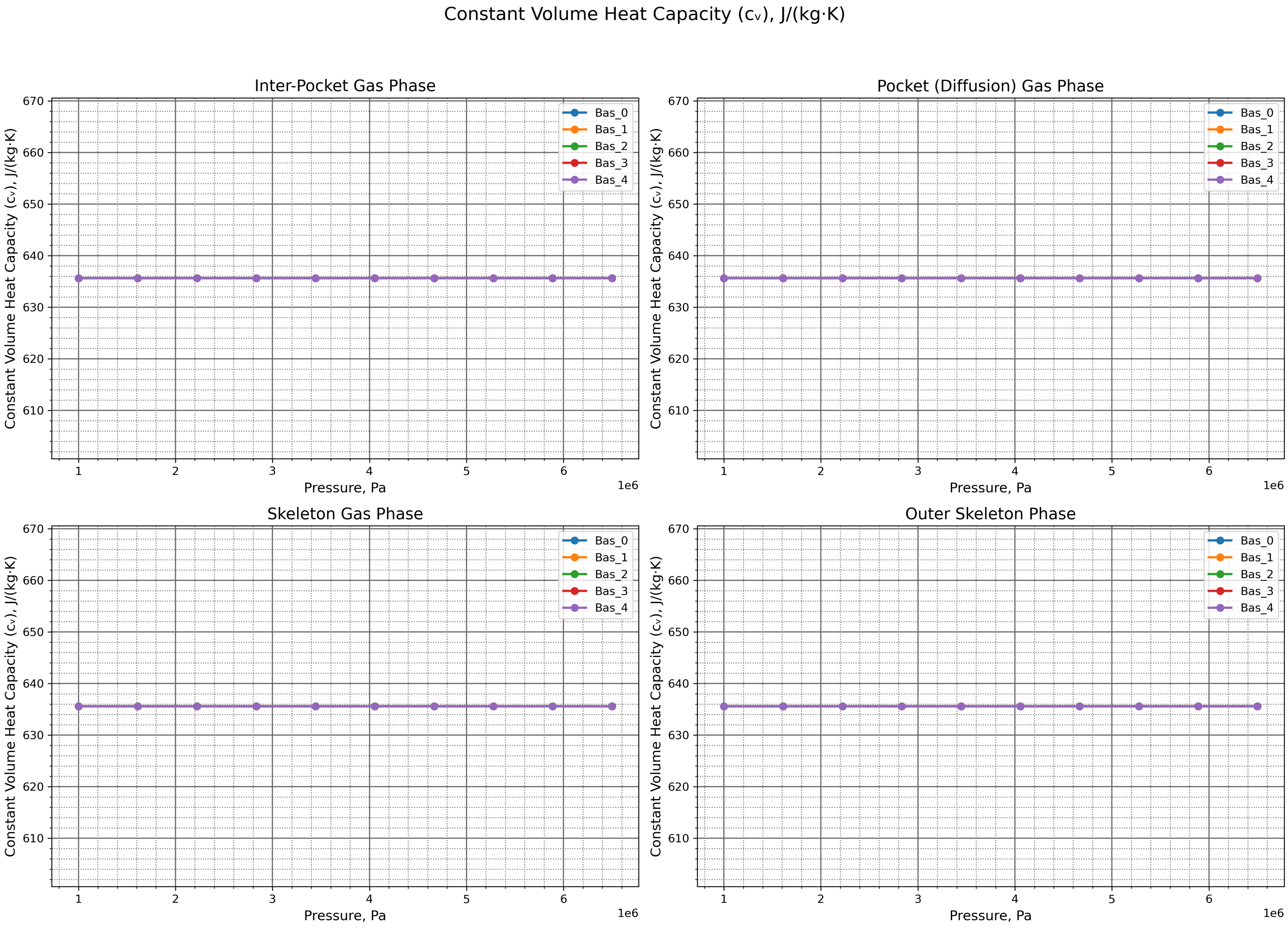
Gas Phase Parameters Outside Skeleton Structure

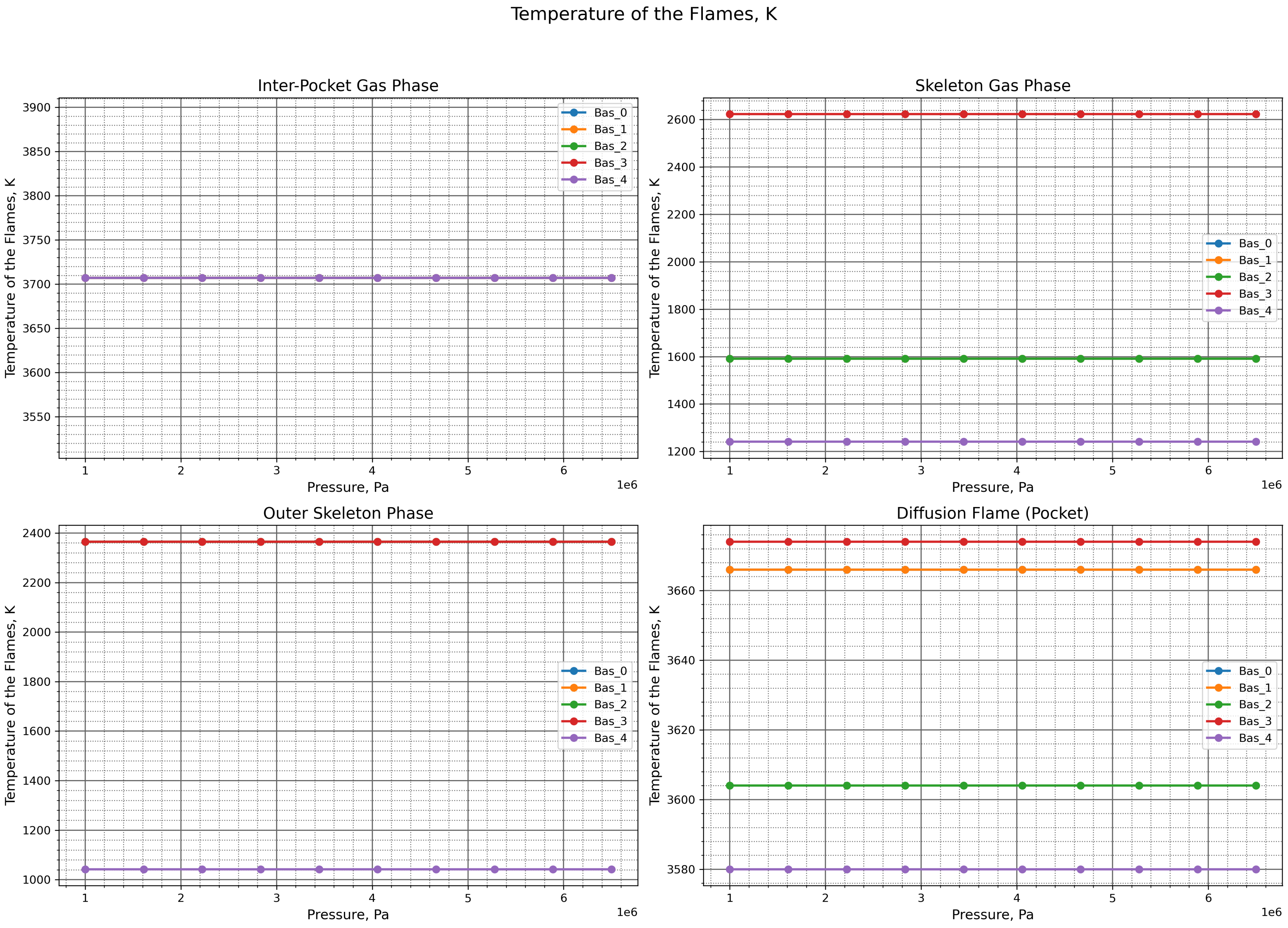
Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

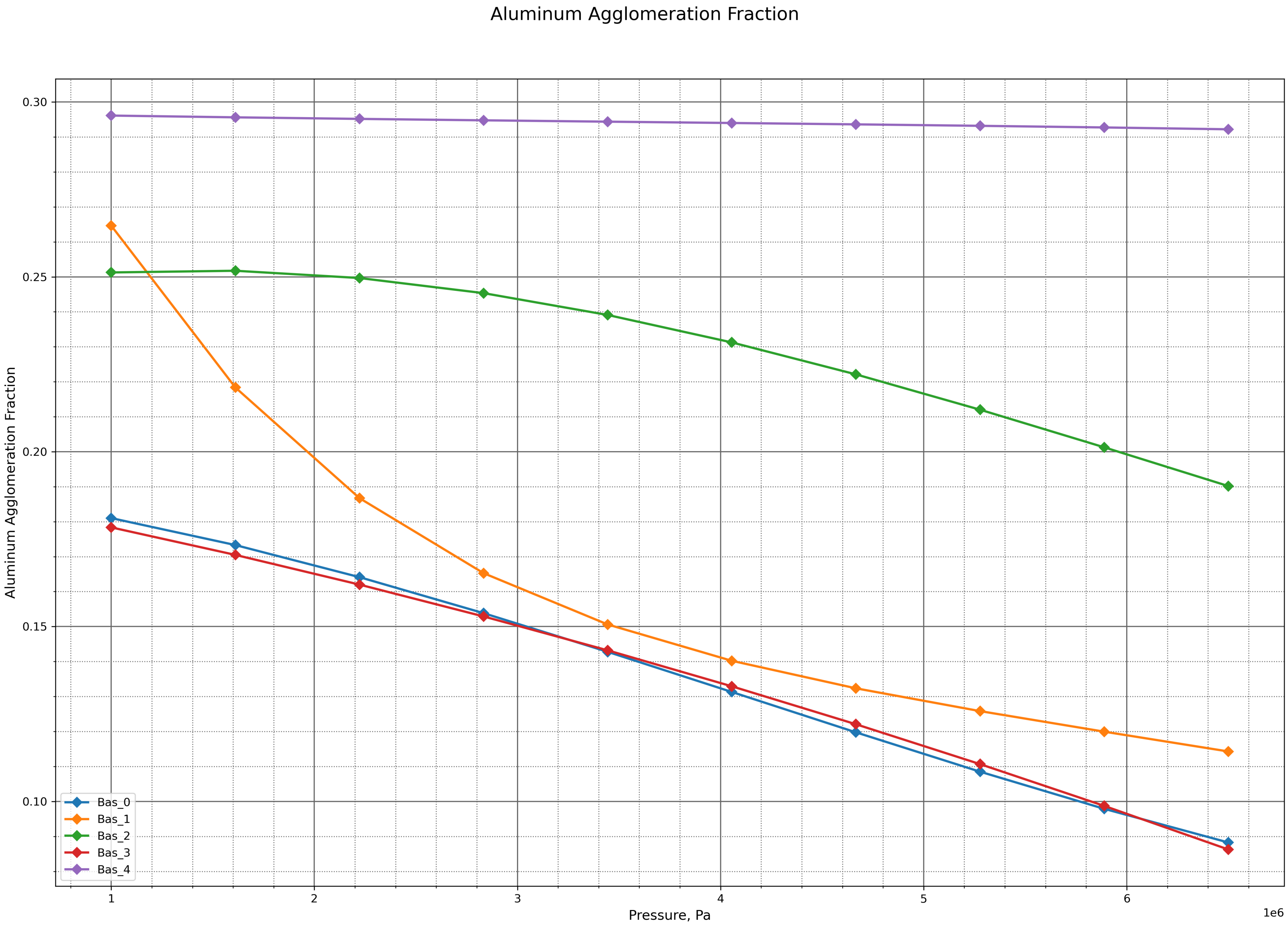


Thermal Conductivity (λ), W/(m·K)
Inter-Pocket Gas Phase
Pocket (Diffusion) Gas Phase
Skeleton Gas Phase
Outer Skeleton Phase
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4
Thermal Conductivity (λ), W/(m·K)
Pressure, Pa
1e6
0.096
0.098
0.100
0.102
0.104
1
2
3
4
5
6

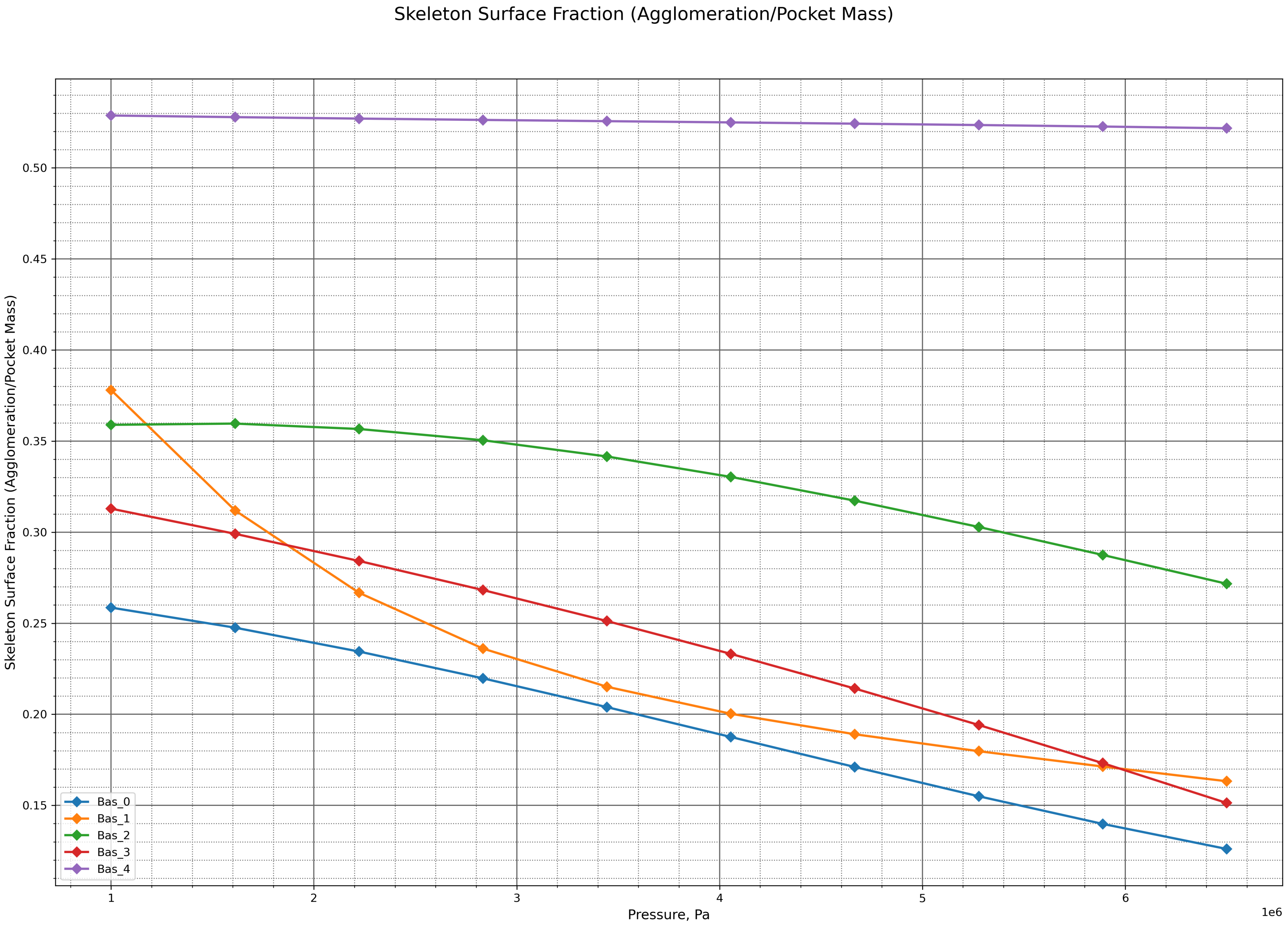

Average Molar Mass, kg/mol
Inter-Pocket Gas Phase
Pocket (Diffusion) Gas Phase
Skeleton Gas Phase
Outer Skeleton Phase
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4
Average Molar Mass, kg/mol
Pressure, Pa
1e6
0.0315
0.0320
0.0325
0.0330
0.0335
0.0340
0.0345
1
2
3
4
5
6

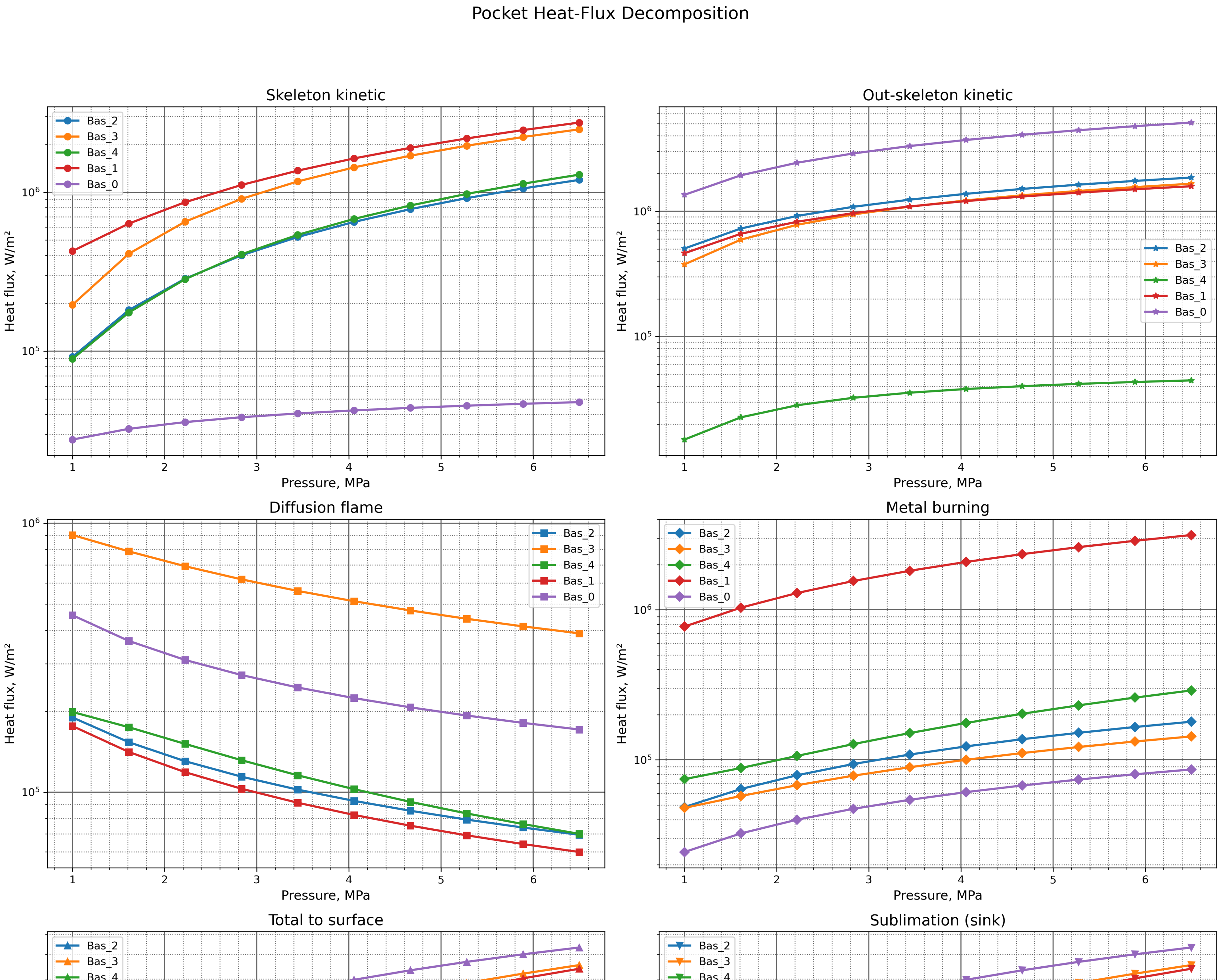

Constant Volume Heat Capacity (cᵥ), J/(kg·K)
Inter-Pocket Gas Phase
Pocket (Diffusion) Gas Phase
Skeleton Gas Phase
Outer Skeleton Phase
Constant Volume Heat Capacity (cᵥ), J/(kg·K)
Pressure, Pa
1e6
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4
670
660
650
640
630
620
610
1
2
3
4
5
6


Temperature of the Flames, K
Inter-Pocket Gas Phase
Skeleton Gas Phase
Outer Skeleton Phase
Diffusion Flame (Pocket)
Temperature of the Flames, K
Pressure, Pa
1e6
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4

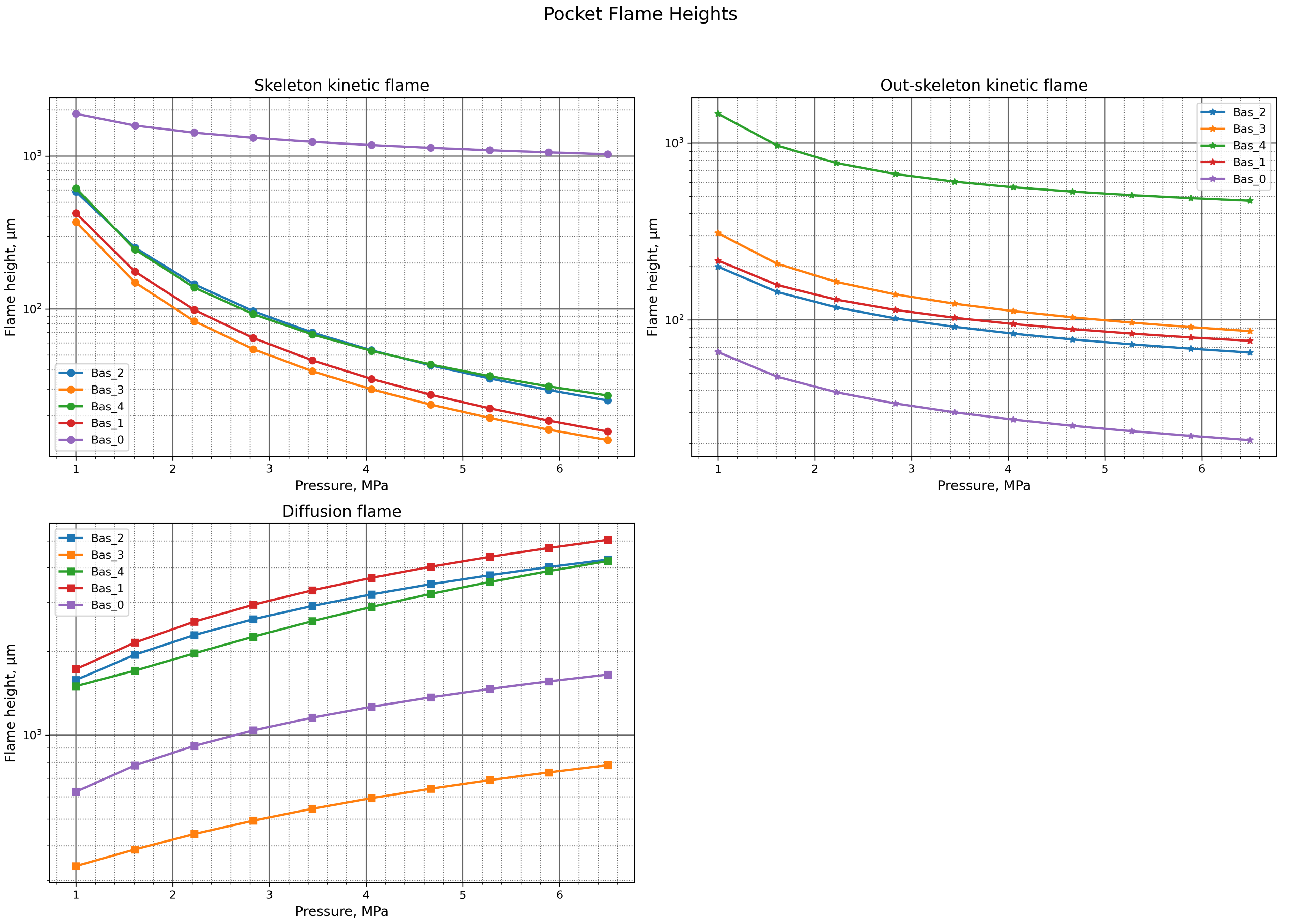


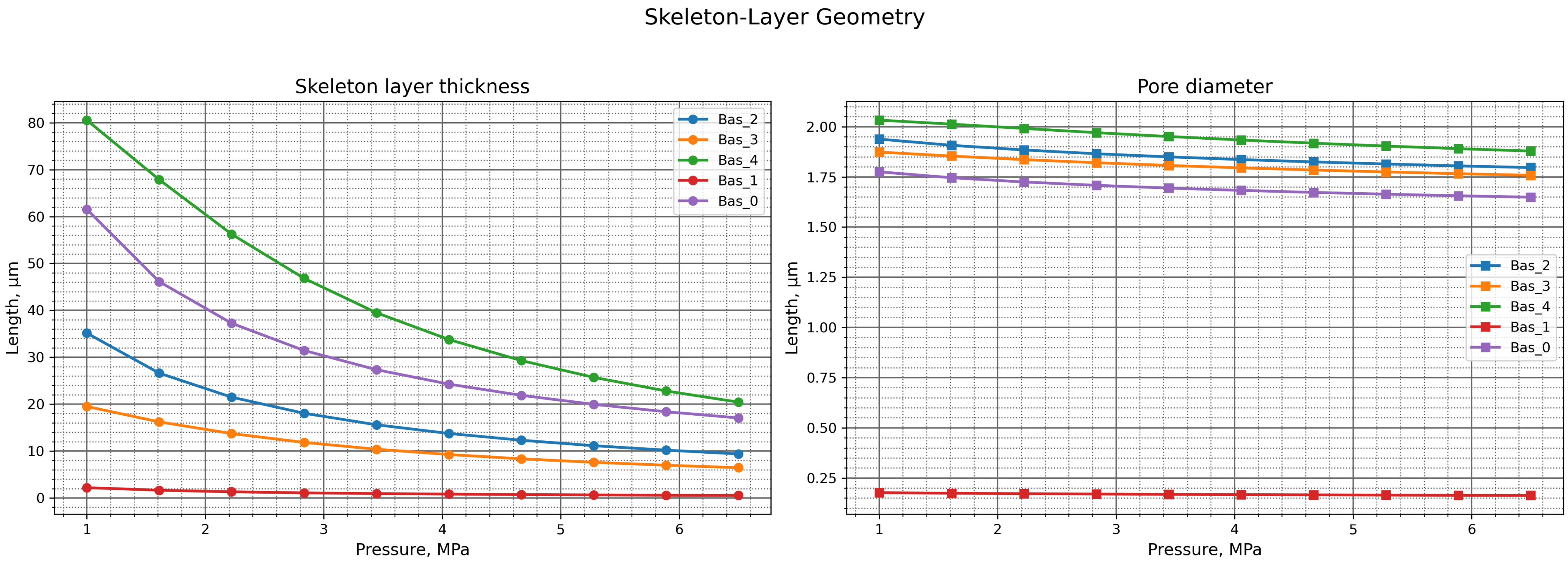
Aluminum Agglomeration Fraction
Aluminum Agglomeration Fraction
Pressure, Pa
1e6
0.30
0.25
0.20
0.15
0.10
1
2
3
4
5
6
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4

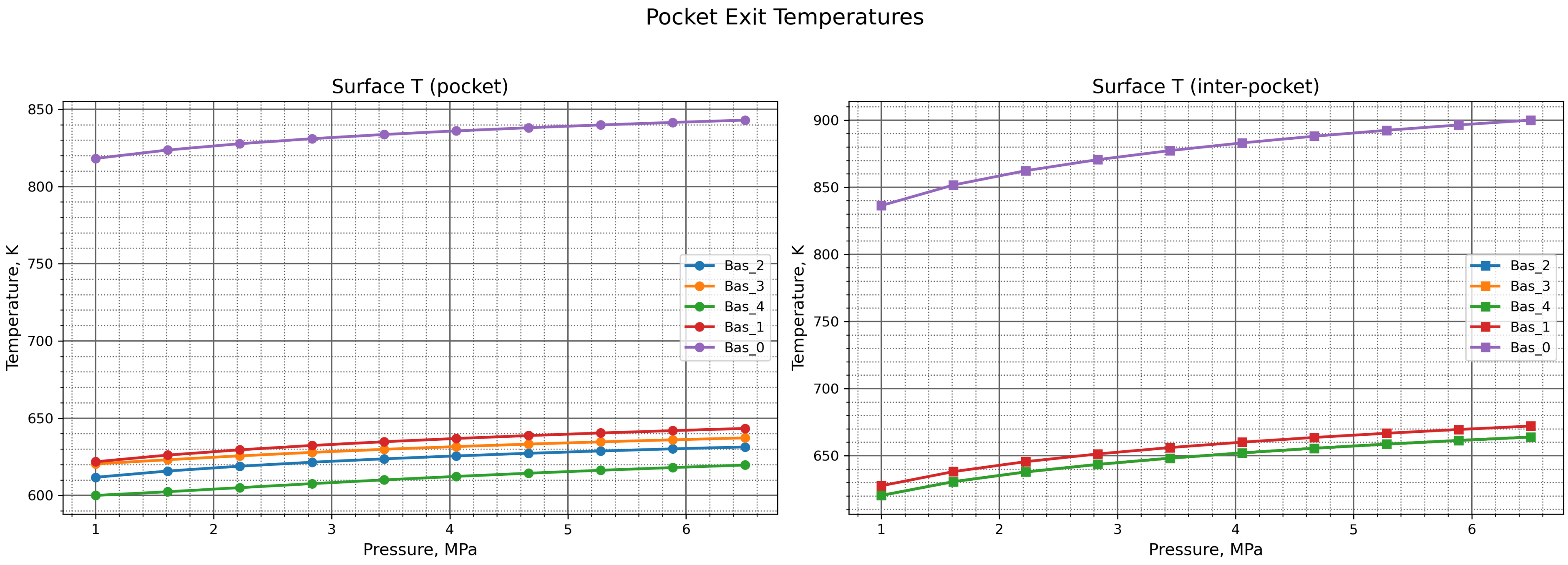

Skeleton Surface Fraction (Agglomeration/Pocket Mass)
Skeleton Surface Fraction (Agglomeration/Pocket Mass)
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.15
1
2
3
4
5
6
1e6
Pressure, Pa
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4



Pocket Heat-Flux Decomposition
Skeleton kinetic
Out-skeleton kinetic
Diffusion flame
Metal burning
Total to surface
Sublimation (sink)
Bas_2
Bas_3
Bas_4
Bas_1
Bas_0
Heat flux, W/m²
Pressure, MPa


Pocket Flame Heights
Skeleton kinetic flame
Out-skeleton kinetic flame
Diffusion flame
Flame height, μm
Pressure, MPa
10^3
10^2
1
2
3
4
5
6
Bas_2
Bas_3
Bas_4
Bas_1
Bas_0


Skeleton-Layer Geometry
Skeleton layer thickness
Length, µm
Pressure, MPa
Bas_2
Bas_3
Bas_4
Bas_1
Bas_0
Pore diameter
Length, µm
Pressure, MPa
Bas_2
Bas_3
Bas_4
Bas_1
Bas_0



Pocket Exit Temperatures
Surface T (pocket)
Surface T (inter-pocket)
Temperature, K
Pressure, MPa
Bas_2
Bas_3
Bas_4
Bas_1
Bas_0
850
800
750
700
650
600
900
1
2
3
4
5
6

Performance Report

System Information
Operating System: Unix 6.8.0.134 (Ubuntu 24.04.4 LTS)
Processor: Intel(R) Xeon(R) CPU E5-2697 v4 @ 2.30GHz (Base Clock Frequency 2.29 GHz)
Memory: 15.6 GB (64-bit), Unknown
Machine Name: vscode
Runtime: .NET 10.0.10 (.NET 10.0.10)
Architecture: X64 / X64

Total Execution Time
Total execution time: 06:08:59.391

Garbage Collection Information
Total Memory: 13374648 bytes
GC Pause Time Percentage: 0.03%
Generation 0 Collections: 3864
Generation 1 Collections: 341
Generation 2 Collections: 38

Detailed Performance Metrics
Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance
Call count: 2590402
Maximum execution time: 64.0818 ms
Minimum execution time: 0.001 ms
Mean execution time: 4.810425910148183 ms
Execution time standard deviation: 0.9701533820677775 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance
Call count: 1954380
Maximum execution time: 73.4266 ms
Minimum execution time: 0.001 ms
Mean execution time: 1.3765706546832883 ms
Execution time standard deviation: 0.7130012599625705 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance
Call count: 1561910
Maximum execution time: 29.852 ms
Minimum execution time: 0.001 ms
Mean execution time: 1.6657114327330649 ms
Execution time standard deviation: 0.27418501911906873 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 43.1241 ms
Minimum execution time: 0.0016 ms
Mean execution time: 4.896745460781217 ms
Execution time standard deviation: 0.9166921929633844 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance
Call count: 1858333
Maximum execution time: 42.6272 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.4036532035970446 ms
Execution time standard deviation: 0.7271008976241795 ms

Method: Telemetry/Proc/1/OptimizationProblemExecutionPerformance
Call count: 1467618
Maximum execution time: 15.9042 ms

Minimum execution time: 0.0013 ms
Mean execution time: 1.7132405882184918 ms
Execution time standard deviation: 0.22199654734337154 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 34.8506 ms
Minimum execution time: 0.0015 ms
Mean execution time: 4.869841461404339 ms
Execution time standard deviation: 0.9335002844445963 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 1858320
Maximum execution time: 23.0735 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.3906877965581923 ms
Execution time standard deviation: 0.7241279482628156 ms

Method: Telemetry/Proc/2/OptimizationProblemExecutionPerformance
Call count: 1466817
Maximum execution time: 26.7024 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.7024893015283948 ms
Execution time standard deviation: 0.2348754278624883 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 60.5398 ms
Minimum execution time: 0.0013 ms
Mean execution time: 4.8869925915303005 ms
Execution time standard deviation: 0.9255178027572377 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 1857574
Maximum execution time: 34.7992 ms
Minimum execution time: 0.0013 ms
Mean execution time: 1.3962895629999303 ms
Execution time standard deviation: 0.7260031478753051 ms

Method: Telemetry/Proc/3/OptimizationProblemExecutionPerformance
Call count: 1466191
Maximum execution time: 20.7992 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.709485698861915 ms
Execution time standard deviation: 0.22764709296529437 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 52.8251 ms
Minimum execution time: 0.0013 ms
Mean execution time: 4.898530604645488 ms
Execution time standard deviation: 0.9239888240699842 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 1860379
Maximum execution time: 18.0248 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.3993096591609406 ms
Execution time standard deviation: 0.7243905642336551 ms

Method: Telemetry/Proc/4/OptimizationProblemExecutionPerformance
Call count: 1470852
Maximum execution time: 21.5941 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.711039774770016 ms
Execution time standard deviation: 0.22924621447423796 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 36.0112 ms
Minimum execution time: 0.0013 ms
Mean execution time: 4.890288606089713 ms
Execution time standard deviation: 0.923367495568652 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance
Call count: 1858576
Maximum execution time: 19.9545 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.395837483266772 ms
Execution time standard deviation: 0.7250743423646868 ms

Method: Telemetry/Proc/5/OptimizationProblemExecutionPerformance
Call count: 1467365
Maximum execution time: 29.4576 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.708684756417156 ms
Execution time standard deviation: 0.23175604961671245 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 42.4395 ms
Minimum execution time: 0.0014 ms
Mean execution time: 4.855068344907268 ms
Execution time standard deviation: 0.9417710447719486 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance
Call count: 1858444
Maximum execution time: 16.2197 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.3870709833602435 ms
Execution time standard deviation: 0.7226826640206362 ms

Method: Telemetry/Proc/6/OptimizationProblemExecutionPerformance
Call count: 1467811
Maximum execution time: 16.1115 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.696970666318804 ms
Execution time standard deviation: 0.23913951887533996 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 34.4688 ms
Minimum execution time: 0.0014 ms
Mean execution time: 4.880413008737434 ms
Execution time standard deviation: 0.9204018826124448 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance
Call count: 1858280

Maximum execution time: 17.227 ms
Minimum execution time: 0.0013 ms
Mean execution time: 1.3971143406806195 ms
Execution time standard deviation: 0.7229725150554305 ms

Method: Telemetry/Proc/7/OptimizationProblemExecutionPerformance
Call count: 1468566
Maximum execution time: 23.2249 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.7078647234103919 ms
Execution time standard deviation: 0.2253684174203626 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 46.855 ms
Minimum execution time: 0.0013 ms
Mean execution time: 4.8835844425356285 ms
Execution time standard deviation: 0.9217451174929031 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance
Call count: 1859531
Maximum execution time: 31.4766 ms
Minimum execution time: 0.0013 ms
Mean execution time: 1.3948699975961603 ms
Execution time standard deviation: 0.7249242703532212 ms

Method: Telemetry/Proc/8/OptimizationProblemExecutionPerformance
Call count: 1468309
Maximum execution time: 21.6802 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.705858999570338 ms
Execution time standard deviation: 0.22750379819321057 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 50.7361 ms
Minimum execution time: 0.0013 ms
Mean execution time: 4.871339969068212 ms
Execution time standard deviation: 0.9275272093144621 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance
Call count: 1859538
Maximum execution time: 20.4614 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.3914279130085099 ms
Execution time standard deviation: 0.7232079932849373 ms

Method: Telemetry/Proc/9/OptimizationProblemExecutionPerformance
Call count: 1468395
Maximum execution time: 32.5799 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.702594078909363 ms
Execution time standard deviation: 0.23602868280072392 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 60.5482 ms
Minimum execution time: 0.0013 ms
Mean execution time: 4.886969456840742 ms

Execution time standard deviation: 0.9232130095761429 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance
Call count: 1857225
Maximum execution time: 18.714 ms
Minimum execution time: 0.0013 ms
Mean execution time: 1.3948541017917166 ms
Execution time standard deviation: 0.7243332414378637 ms

Method: Telemetry/Proc/10/OptimizationProblemExecutionPerformance
Call count: 1466245
Maximum execution time: 18.2722 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.7086824399060974 ms
Execution time standard deviation: 0.22663311437875805 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 43.5117 ms
Minimum execution time: 0.0013 ms
Mean execution time: 4.853452854953737 ms
Execution time standard deviation: 0.9353551507937571 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance
Call count: 1858558
Maximum execution time: 22.2019 ms
Minimum execution time: 0.0013 ms
Mean execution time: 1.3825573270245362 ms
Execution time standard deviation: 0.7208241187024953 ms

Method: Telemetry/Proc/11/OptimizationProblemExecutionPerformance
Call count: 1467879
Maximum execution time: 37.8339 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.692968896346325 ms
Execution time standard deviation: 0.24258702141974867 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 32.5535 ms
Minimum execution time: 0.0014 ms
Mean execution time: 4.863604091987659 ms
Execution time standard deviation: 0.9350353399862974 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance
Call count: 1858539
Maximum execution time: 56.1384 ms
Minimum execution time: 0.0013 ms
Mean execution time: 1.3883132742438717 ms
Execution time standard deviation: 0.7257644664981425 ms

Method: Telemetry/Proc/12/OptimizationProblemExecutionPerformance
Call count: 1467555
Maximum execution time: 18.6853 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.699029522028143 ms
Execution time standard deviation: 0.2384534835733585 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance

Call count: 2489988
Maximum execution time: 57.1693 ms
Minimum execution time: 0.0015 ms
Mean execution time: 4.889830609625422 ms
Execution time standard deviation: 0.9232637374956819 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance
Call count: 1858165
Maximum execution time: 21.5338 ms
Minimum execution time: 0.0016 ms
Mean execution time: 1.398627937992621 ms
Execution time standard deviation: 0.7238649815174921 ms

Method: Telemetry/Proc/13/OptimizationProblemExecutionPerformance
Call count: 1468823
Maximum execution time: 34.5282 ms
Minimum execution time: 0.0015 ms
Mean execution time: 1.7092932288642497 ms
Execution time standard deviation: 0.2274046569148486 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 45.0148 ms
Minimum execution time: 0.0014 ms
Mean execution time: 4.879445849899873 ms
Execution time standard deviation: 0.9293742182313085 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance
Call count: 1858861
Maximum execution time: 33.7749 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.3920296575160502 ms
Execution time standard deviation: 0.7255120787274927 ms

Method: Telemetry/Proc/14/OptimizationProblemExecutionPerformance
Call count: 1467316
Maximum execution time: 13.147 ms
Minimum execution time: 0.0013 ms
Mean execution time: 1.7044548420381314 ms
Execution time standard deviation: 0.23229012760926693 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 31.1905 ms
Minimum execution time: 0.0013 ms
Mean execution time: 4.900134034501499 ms
Execution time standard deviation: 0.9176837272653285 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance
Call count: 1858501
Maximum execution time: 19.4389 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.4014432818707763 ms
Execution time standard deviation: 0.724997587052068 ms

Method: Telemetry/Proc/15/OptimizationProblemExecutionPerformance
Call count: 1468866
Maximum execution time: 37.6035 ms
Minimum execution time: 0.0014 ms

Mean execution time: 1.7126674192880944 ms
Execution time standard deviation: 0.22752225500035383 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 36.252 ms
Minimum execution time: 0.0014 ms
Mean execution time: 4.876390076498312 ms
Execution time standard deviation: 0.9255679562379925 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 1857862
Maximum execution time: 32.8935 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.3927732870902132 ms
Execution time standard deviation: 0.7232265226888472 ms

Method: Telemetry/Proc/16/OptimizationProblemExecutionPerformance
Call count: 1466986
Maximum execution time: 23.8596 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.7088688133356487 ms
Execution time standard deviation: 0.22823697998597084 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 50.5902 ms
Minimum execution time: 0.0013 ms
Mean execution time: 4.886056560714628 ms
Execution time standard deviation: 0.9324693258637711 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 1858945
Maximum execution time: 34.3149 ms
Minimum execution time: 0.0013 ms
Mean execution time: 1.392837257960779 ms
Execution time standard deviation: 0.7241152435539452 ms

Method: Telemetry/Proc/17/OptimizationProblemExecutionPerformance
Call count: 1469047
Maximum execution time: 25.3093 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.7038953970159307 ms
Execution time standard deviation: 0.23463501021024968 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 38.812 ms
Minimum execution time: 0.0013 ms
Mean execution time: 4.895255959064788 ms
Execution time standard deviation: 0.9182091234545023 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 1857873
Maximum execution time: 23.5792 ms
Minimum execution time: 0.0013 ms
Mean execution time: 1.3975340343500722 ms
Execution time standard deviation: 0.7255282250242137 ms

Method: Telemetry/Proc/18/OptimizationProblemExecutionPerformance
Call count: 1466664
Maximum execution time: 42.1072 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.7112159998472225 ms
Execution time standard deviation: 0.2293215147832189 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 40.6331 ms
Minimum execution time: 0.0013 ms
Mean execution time: 4.8757416882330675 ms
Execution time standard deviation: 0.9300448030439876 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 1857426
Maximum execution time: 22.0683 ms
Minimum execution time: 0.0013 ms
Mean execution time: 1.3932972144247082 ms
Execution time standard deviation: 0.7239434566327582 ms

Method: Telemetry/Proc/19/OptimizationProblemExecutionPerformance
Call count: 1467797
Maximum execution time: 16.1265 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.7035142338484532 ms
Execution time standard deviation: 0.2334880102965105 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 67.2447 ms
Minimum execution time: 0.0013 ms
Mean execution time: 4.8852257251838695 ms
Execution time standard deviation: 0.9237384044277408 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 1858414
Maximum execution time: 25.6576 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.3965153915113326 ms
Execution time standard deviation: 0.7240365532819701 ms

Method: Telemetry/Proc/20/OptimizationProblemExecutionPerformance
Call count: 1468823
Maximum execution time: 31.4395 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.706926183345491 ms
Execution time standard deviation: 0.23162648892489324 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 64.3774 ms
Minimum execution time: 0.0013 ms
Mean execution time: 4.8816452413429126 ms
Execution time standard deviation: 0.9257643887901287 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 1857373
Maximum execution time: 14.5951 ms

Minimum execution time: 0.0013 ms
Mean execution time: 1.3943628652940805 ms
Execution time standard deviation: 0.724043700137334 ms

Method: Telemetry/Proc/21/OptimizationProblemExecutionPerformance
Call count: 1465589
Maximum execution time: 30.6144 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.707569128998661 ms
Execution time standard deviation: 0.2301933837833344 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 53.2539 ms
Minimum execution time: 0.0013 ms
Mean execution time: 4.890273467502393 ms
Execution time standard deviation: 0.9184599460319927 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 1858068
Maximum execution time: 14.9432 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.3996640692913838 ms
Execution time standard deviation: 0.7236621349156036 ms

Method: Telemetry/Proc/22/OptimizationProblemExecutionPerformance
Call count: 1468731
Maximum execution time: 29.8188 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.710251906577746 ms
Execution time standard deviation: 0.22729529775174429 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 42.141 ms
Minimum execution time: 0.0013 ms
Mean execution time: 4.903416029193397 ms
Execution time standard deviation: 0.9225591732677327 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 1857800
Maximum execution time: 28.8411 ms
Minimum execution time: 0.0013 ms
Mean execution time: 1.4005102450748461 ms
Execution time standard deviation: 0.7261746952052265 ms

Method: Telemetry/Proc/23/OptimizationProblemExecutionPerformance
Call count: 1467616
Maximum execution time: 19.0911 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.7156982464758006 ms
Execution time standard deviation: 0.22540380233081042 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 49.3461 ms
Minimum execution time: 0.0013 ms
Mean execution time: 4.910151726193029 ms
Execution time standard deviation: 0.9171731026432353 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance
Call count: 1858571
Maximum execution time: 27.8838 ms
Minimum execution time: 0.0013 ms
Mean execution time: 1.4012284988305839 ms
Execution time standard deviation: 0.7260822788403049 ms

Method: Telemetry/Proc/24/OptimizationProblemExecutionPerformance
Call count: 1467531
Maximum execution time: 30.2722 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.7156189480154012 ms
Execution time standard deviation: 0.22486780554308883 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 61.0896 ms
Minimum execution time: 0.0013 ms
Mean execution time: 4.889475024779123 ms
Execution time standard deviation: 0.9262208229371679 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance
Call count: 1857837
Maximum execution time: 26.1623 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.3964615418360011 ms
Execution time standard deviation: 0.7242795567274777 ms

Method: Telemetry/Proc/25/OptimizationProblemExecutionPerformance
Call count: 1467542
Maximum execution time: 29.9415 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.7085801401255662 ms
Execution time standard deviation: 0.23304231650079665 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 41.3922 ms
Minimum execution time: 0.0013 ms
Mean execution time: 4.896940576861876 ms
Execution time standard deviation: 0.9192514856617323 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance
Call count: 1858723
Maximum execution time: 23.8527 ms
Minimum execution time: 0.0015 ms
Mean execution time: 1.3994483283415713 ms
Execution time standard deviation: 0.7248167801914215 ms

Method: Telemetry/Proc/26/OptimizationProblemExecutionPerformance
Call count: 1468687
Maximum execution time: 26.7482 ms
Minimum execution time: 0.0015 ms
Mean execution time: 1.71201306112189 ms
Execution time standard deviation: 0.22652659715876083 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance
Call count: 2489988

Maximum execution time: 46.4238 ms
Minimum execution time: 0.0013 ms
Mean execution time: 4.879342518518118 ms
Execution time standard deviation: 0.9266992704755036 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance
Call count: 1857926
Maximum execution time: 24.4224 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.396929501067364 ms
Execution time standard deviation: 0.724666703816992 ms

Method: Telemetry/Proc/27/OptimizationProblemExecutionPerformance
Call count: 1468636
Maximum execution time: 23.9877 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.705331828853468 ms
Execution time standard deviation: 0.23199885777796728 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 62.872 ms
Minimum execution time: 0.0015 ms
Mean execution time: 4.883378498932419 ms
Execution time standard deviation: 0.9292806203910333 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance
Call count: 1858666
Maximum execution time: 26.4118 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.3977584980840547 ms
Execution time standard deviation: 0.7254589853641612 ms

Method: Telemetry/Proc/28/OptimizationProblemExecutionPerformance
Call count: 1468174
Maximum execution time: 33.2364 ms
Minimum execution time: 0.0015 ms
Mean execution time: 1.7071338776603073 ms
Execution time standard deviation: 0.23416218490310217 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 49.7599 ms
Minimum execution time: 0.0013 ms
Mean execution time: 4.873229358334549 ms
Execution time standard deviation: 0.930274185504849 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance
Call count: 1858866
Maximum execution time: 19.01 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.392647538230344 ms
Execution time standard deviation: 0.722449288653952 ms

Method: Telemetry/Proc/29/OptimizationProblemExecutionPerformance
Call count: 1468829
Maximum execution time: 45.1218 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.709384415204179 ms

Execution time standard deviation: 0.23502484648469635 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 70.7245 ms
Minimum execution time: 0.0013 ms
Mean execution time: 4.875818907440235 ms
Execution time standard deviation: 0.934565172890958 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance
Call count: 1857500
Maximum execution time: 16.78 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.391826888613743 ms
Execution time standard deviation: 0.7231767856082318 ms

Method: Telemetry/Proc/30/OptimizationProblemExecutionPerformance
Call count: 1467129
Maximum execution time: 16.5574 ms
Minimum execution time: 0.0013 ms
Mean execution time: 1.703214902643239 ms
Execution time standard deviation: 0.23197515424627393 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance
Call count: 2489988
Maximum execution time: 44.5367 ms
Minimum execution time: 0.0013 ms
Mean execution time: 4.864147465570153 ms
Execution time standard deviation: 0.9189962053252085 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance
Call count: 1856611
Maximum execution time: 26.1485 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.3903112527072101 ms
Execution time standard deviation: 0.7211540035958826 ms

Method: Telemetry/Proc/31/OptimizationProblemExecutionPerformance
Call count: 1465849
Maximum execution time: 24.1276 ms
Minimum execution time: 0.0014 ms
Mean execution time: 1.7020307527582932 ms
Execution time standard deviation: 0.22756829713575819 ms